Machine Learning and Data Mining:

Neural Network

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Some of the materials in this lecture are taken from the Machine Learning Course by Andrew Ng on Coursera.com.

Motivation

Neural Networks in Bioinformatics

- High-dimensional biological data
- nonlinear biological mechanisms
- predictive modeling challenges

Examples of Bioinformatics Problems

- Cancer classification from gene expression
- Metabolomics disease prediction
- Protein function prediction
- Microbiome community analysis

Key message: Biology is rarely linear.

Why Classical Models May Fail

Not able to handle:

- Interaction effects
- Threshold behavior
- Pathway-level nonlinear regulation
- Measurement noise

Multilayer Perceptron

Neural networks include many different architectures. In this lecture, we focus on **multilayer perceptrons (MLPs)**, a class of feed-forward neural networks consisting of fully connected layers. MLPs provide the foundation for understanding more advanced architectures such as CNNs, RNNs, and Transformers.

Introduction

ANN as Nonlinear Function Approximator



A Neural Network is a mathematical system that learns:

$$y = f(x)$$

where

- x = biological measurements (gene expression, metabolites, SNPs, microbiome abundance etc)
- y = biological outcome (disease status, survival, phenotype, etc)

Neural networks can learn very complex nonlinear biological relationships that classical models cannot represent.

Why "Nonlinear" Matter?

Biological systems are inherently nonlinear.

- Example A - Gene regulation threshold effect
  - Gene A has no effect until expression exceeds a threshold
  - After threshold → strong activation of disease pathway

  This cannot be modeled well by linear regression.

- Example B — Interaction between metabolites. Disease risk may increase only when:
  - Glucose is high AND
  - triglycerides are high

  This is an interaction surface, not a straight line.

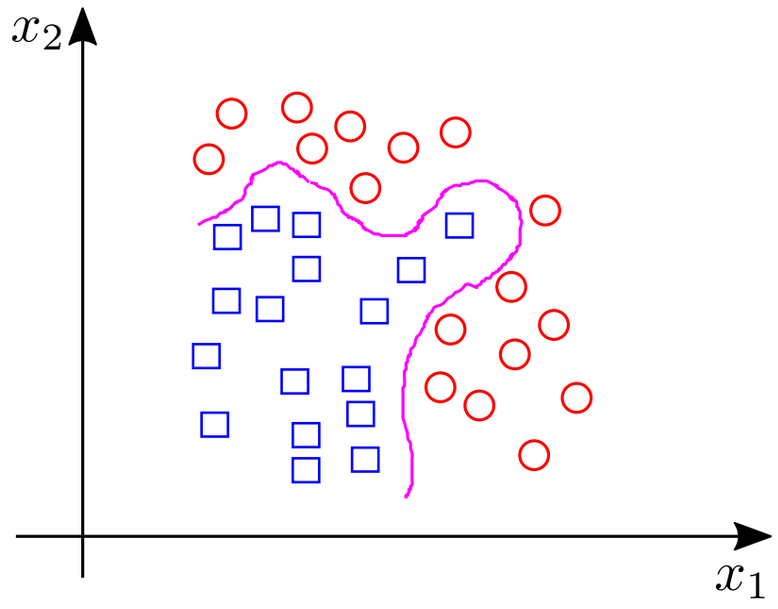
- Example C — Saturation effects. Biological receptors often follow:
  - saturation kinetics
  - sigmoid-like response.

  This is naturally nonlinear.

History of Neural Network

- Origins: algorithms that try to mimic the brain.
- Was very widely used in the 80's and early 90's.
- Popularity diminished in late 90's.
- Recent resurgence: state-of-the-art technique for many application.

Nonlinear Classification



- If logistic regression is used, higher-order terms need to be considered for a complex nonlinear classification:

$$g(z) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1 x_2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^3 x_2 + \theta_6 x_1 x_2^2 + \cdots)$$

- However, the number of higher-order terms increases exponentially with the number of features.

  Quadratic terms, like $x_1^2, x_1 x_2, x_1 x_3, \cdots$, are $\sim \mathcal{O}(n^2)$

  Cubic terms, like $x_1 x_2 x_3, x_1^2 x_2, x_{20} x_{30} x_{40}, \cdots$, are $\sim \mathcal{O}(n^3)$.

- To train a logistic regression model with so many terms is very challenging.

Nonlinear Classification

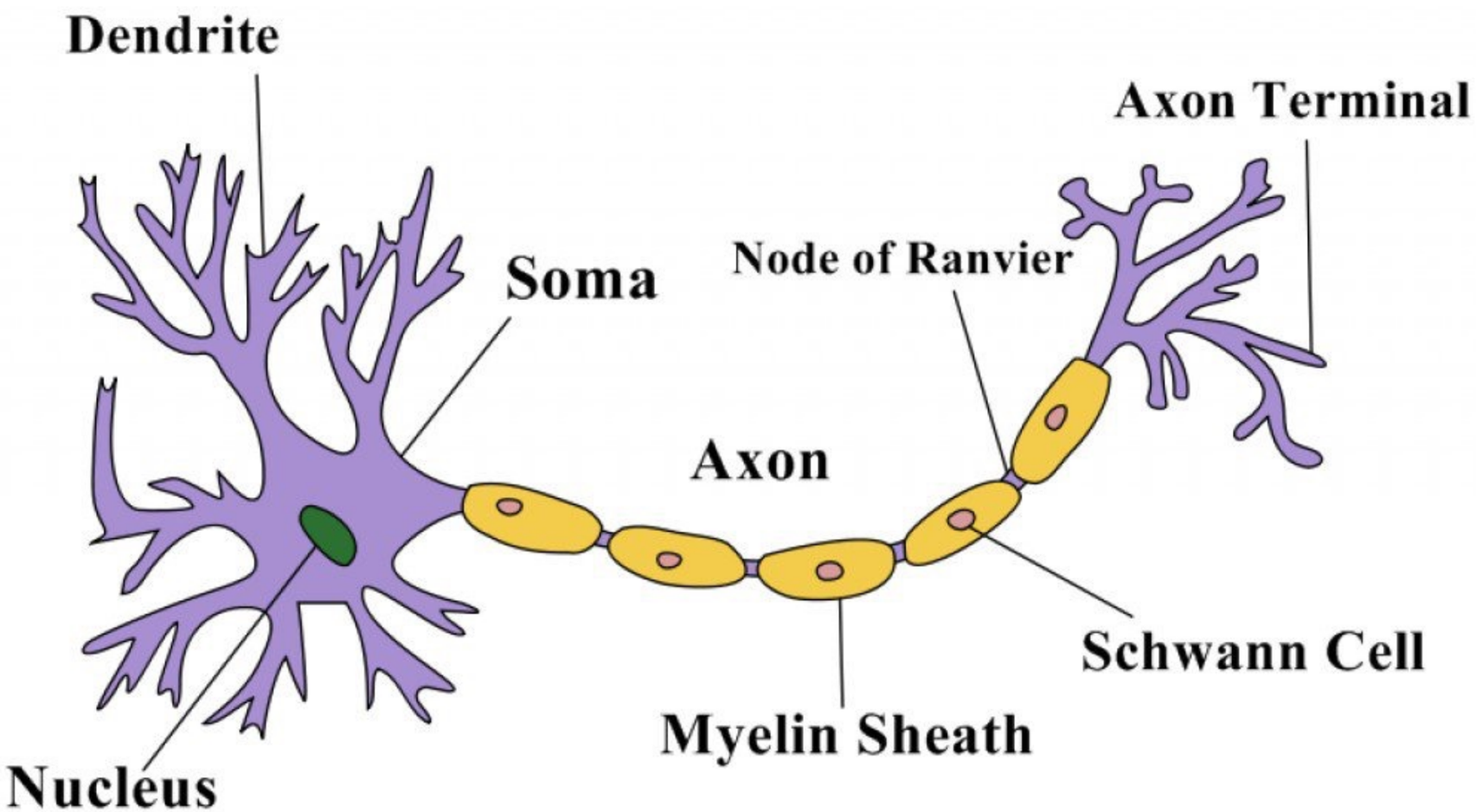
- Example: image recognition

  If the number of pixels in an image is 50×50, then the total number of pixels is 2500. That means that the number of features n is 2500 if gray scale image is used. If pixel value are represented in RGB format, then the number of features n is 7500.

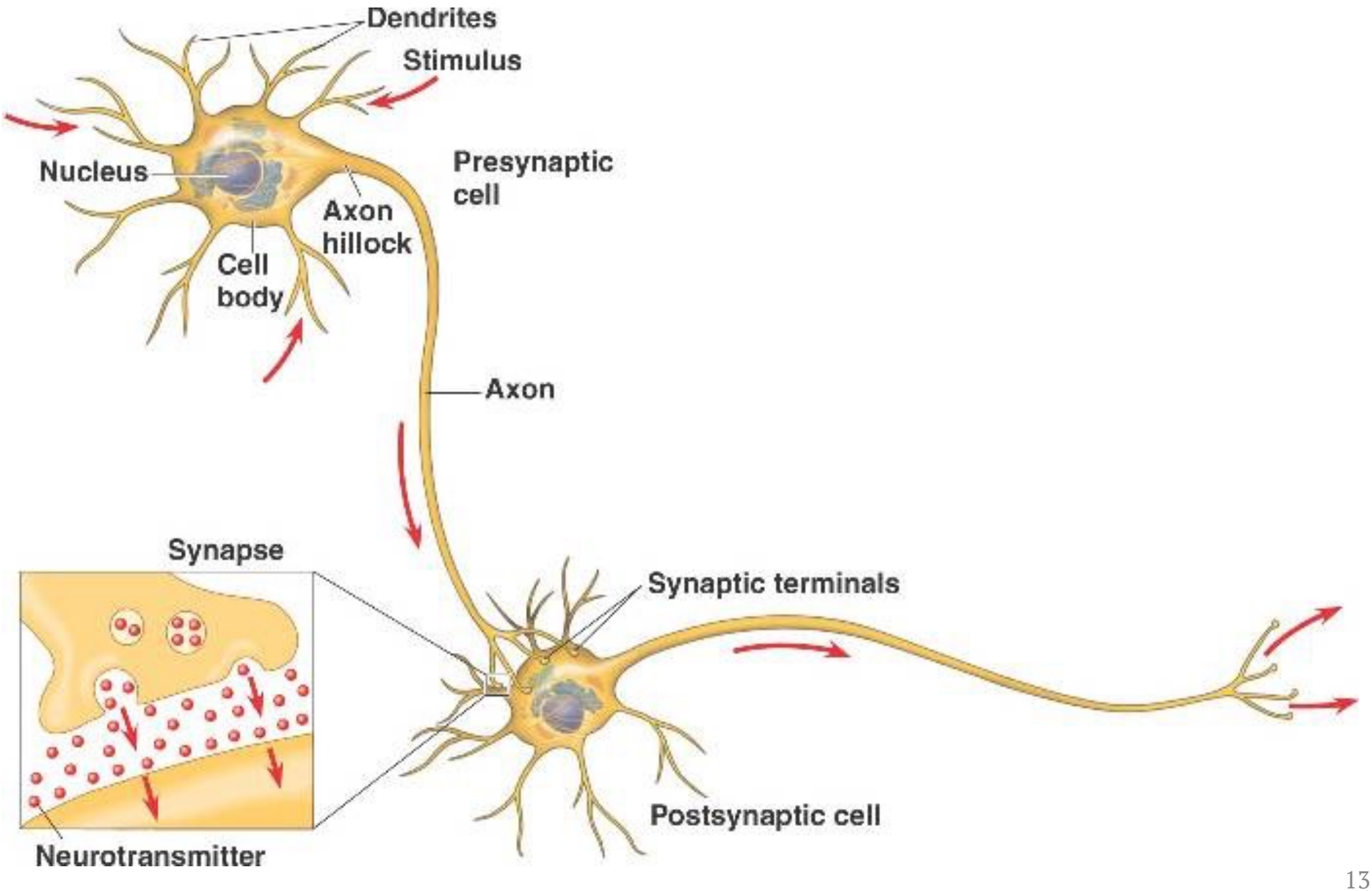
  If quadratic or cubic terms are also considered, using a logistic regression classifier becomes very difficult.

One solution: use artificial neural network that imitates real neural network.

A Real Neuron

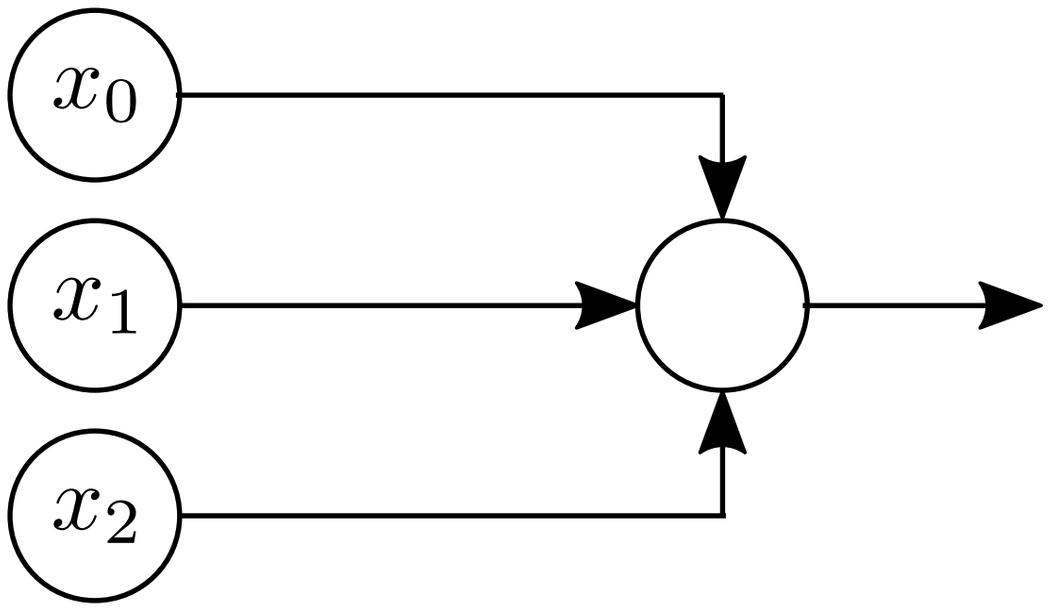


Communications between Real Neurons



Artificial Neurons

Neuron Model



$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2)}}$$

with $x_0 = 1$ representing the bias term and h_θ is the sigmoid/logistic activation function. $\theta_0, \theta_1, \theta_2$ are called weights or parameters.

In general,

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix}, \quad h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Neuron Model

Mathematical framework:

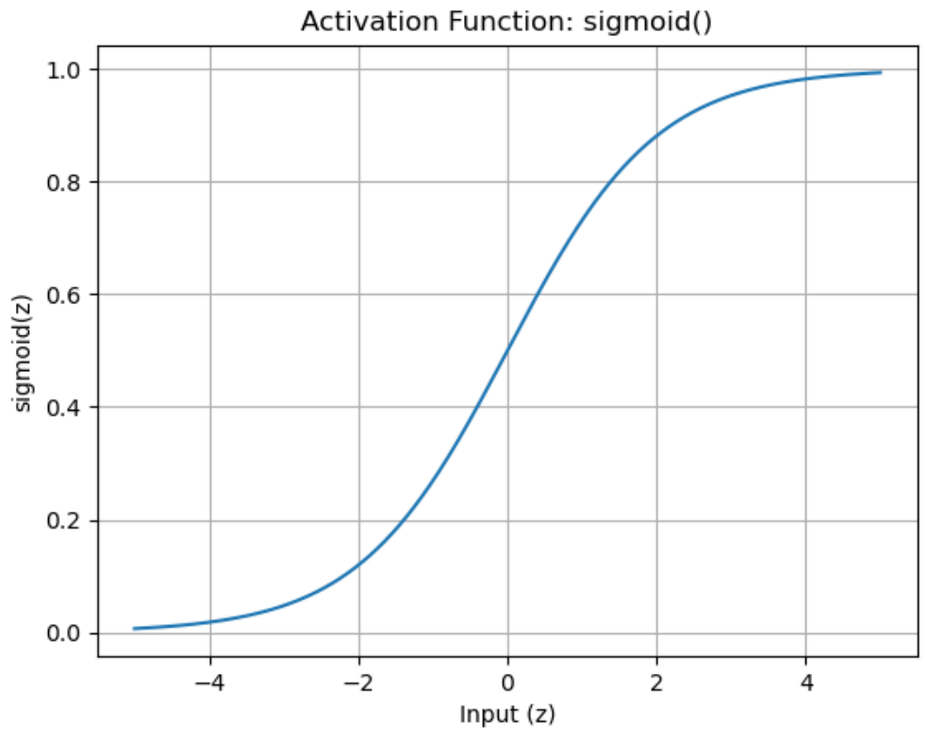
- Linear combination of x:

$$z = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

- Sigmoid activation

$$h_\theta(x) = \frac{1}{1 + e^{-z}}$$

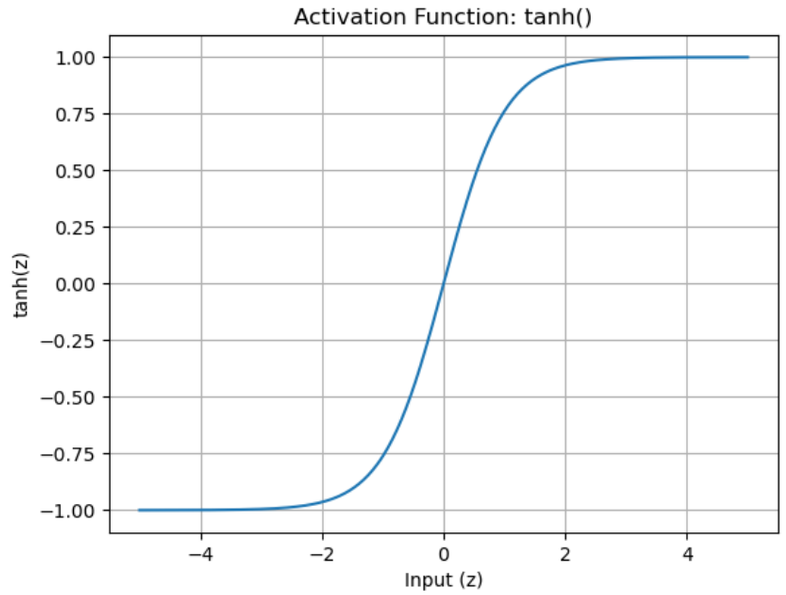
Biological intuition: Sigmoid behaves like biological saturation.



Alternative Activation Function: tanh()

The hyperbolic tangent activation function is:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$



Important properties:

- Output range:

$$-1 \leq \tanh(z) \leq 1$$

- Symmetric around zero
- Smooth and differentiable

This makes it suitable for gradient-based learning.

Interpretation:

- z is the weighted biomarker signal.
- $\tanh(z)$ is the nonlinear transformation from z into activation vs suppression biological signal.

Alternative Activation Function: tanh()

Propertise of tanh():

- For large positive input $\rightarrow$ output approaches +1
- For large negative input $\rightarrow$ output approaches -1
- Near zero $\rightarrow$ approximately linear

Intuition: tanh behaves like a soft biological switch with inhibitory (negative) and activating (positive) states.

Alternative Activation Function: tanh()

Derivative is elegant:

$$\frac{d}{dz}\tanh(z) = 1 - \tanh^2(z)$$

Properties:

- derivative largest near $z = 0$
- derivative very small at extremes

This leads to: **Vanishing gradient problem**, i.e. when neuron saturates near ± 1:

- gradient ≈ 0
- learning slows

This is why modern deep learning prefers ReLU.

tanh() and sigmoid()

Relationship between tanh() and sigmoid():

$$\tanh(z) = 2 \cdot \text{sigmoid}(2z) - 1$$

- sigmoid output range $\rightarrow$ 0 to 1
- tanh output range $\rightarrow$ -1 to 1

Alternative Activation Function: ReLU()

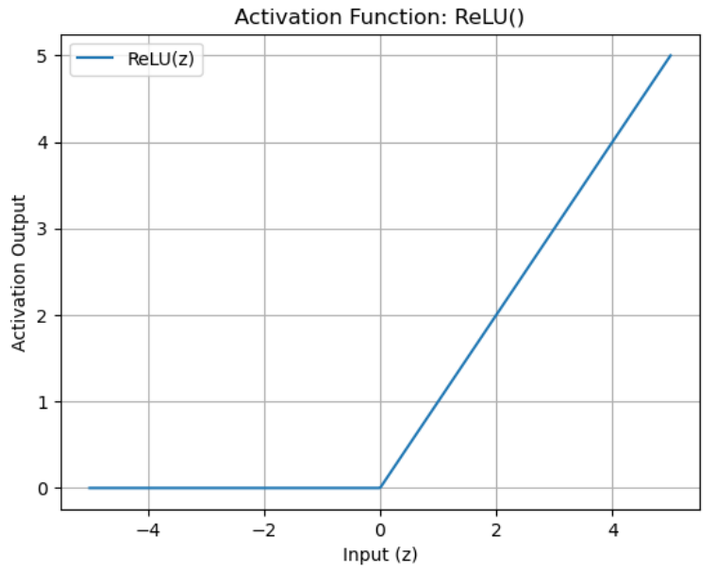
The Rectified Linear Unit (ReLU) activation function is:

$$\text{ReLU}(z) = \max(0, z)$$

This means:

- If $z < 0 \rightarrow$ output $= 0$
- If $z \geq 0 \rightarrow$ output $=$ z

So the function is piecewise linear.



Intuition:

- ReLU behaves like a biological activation threshold.
- The neuron is inactive when signal is weak.
- The neuron is linearly responsive when signal exceeds threshold.

Alternative Activation Function: ReLU()

Historically:

- sigmoid and tanh were widely used.
- deep networks trained very slowly.

Why ReLU became dominant in deep learning? ReLU improved training because:

- It avoids the vanishing gradient problem for positive z

$$\frac{d}{dz}\text{ReLU}(z) = \begin{cases} 0 & \text{if} \quad z < 0 \\ 1 & \text{if} \quad z > 0 \end{cases}$$

  - gradients do not shrink in positive z region.
  - deep networks can learn more efficiently.

- Computational simplicity: no exponentials and fast

Alternative Activation Function: ReLU()

Why ReLU became dominant in deep learning? ReLU improved training because:

- Sparse activation property
  - many neurons output zero.
  - network becomes sparse.

  Interpretation: Neural network selects only a subset of biological signals to represent disease state.

- Biological interpretation:
  - below threshold → pathway inactive
  - above threshold → activity increases proportionally

  This is realistic for enzyme activation, gene regulation, and signaling cascades. Thus ReLU models **pathway "turn-on" dynamics**.

Alternative Activation Function: ReLU()

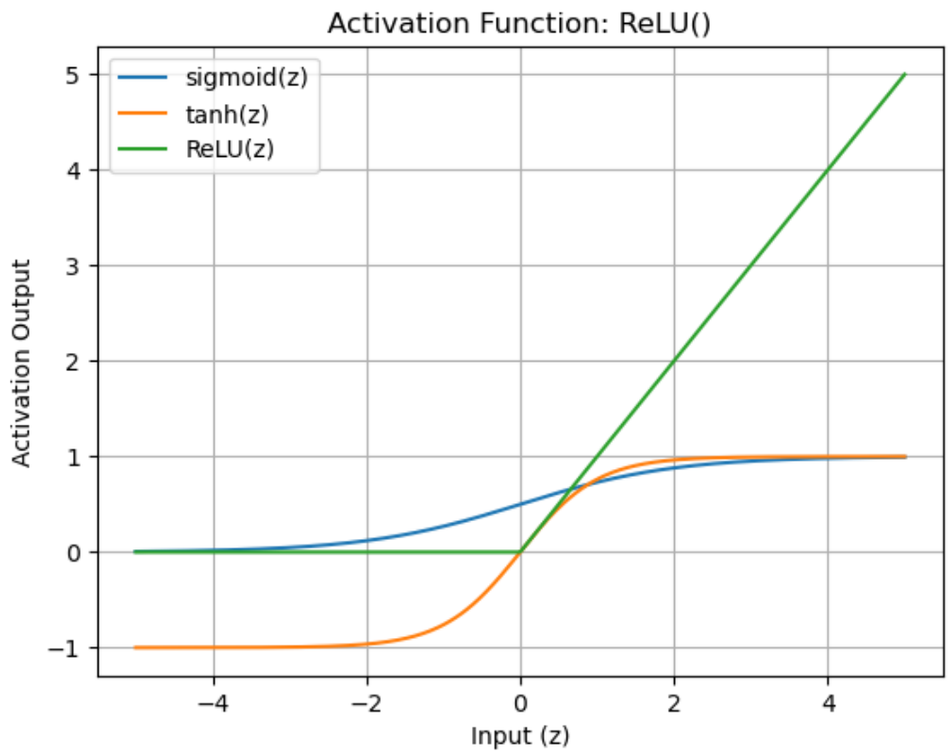
- Important limitation – Dying ReLU problem

  If neuron receives strongly negative inputs, then it may output 0 for all future inputs. Then gradient $= 0$ and neuron never recovers. This is called **dying ReLU**.

Sigmoid, tanh, and ReLU

Function	Output Range	Zero-centered	Training Speed	Modern Use
Sigmoid	0 to 1	No	slow	output layer mainly
tanh	-1 to 1	Yes	moderate	hidden layers
ReLU	0 to ∞	No	fast	dominant in deep learning

tanh is mathematically elegant and pedagogically excellent even if ReLU is more common in modern architectures.



Alternative Activation Function: ReLU()

Modern variants of ReLU:

- Leaky ReLU

$$\text{LeakyReLU}(z) = \begin{cases} 0.01z & \text{for} \quad z < 0 \\ z & \text{for} \quad z \geq 0 \end{cases}$$

- Exponential Linear Unit (ELU)

$$\text{ELU}(z) = \begin{cases} z & \text{for} \quad z \geq 0 \\ \alpha(e^z - 1) & \text{for} \quad z < 0 \end{cases}$$

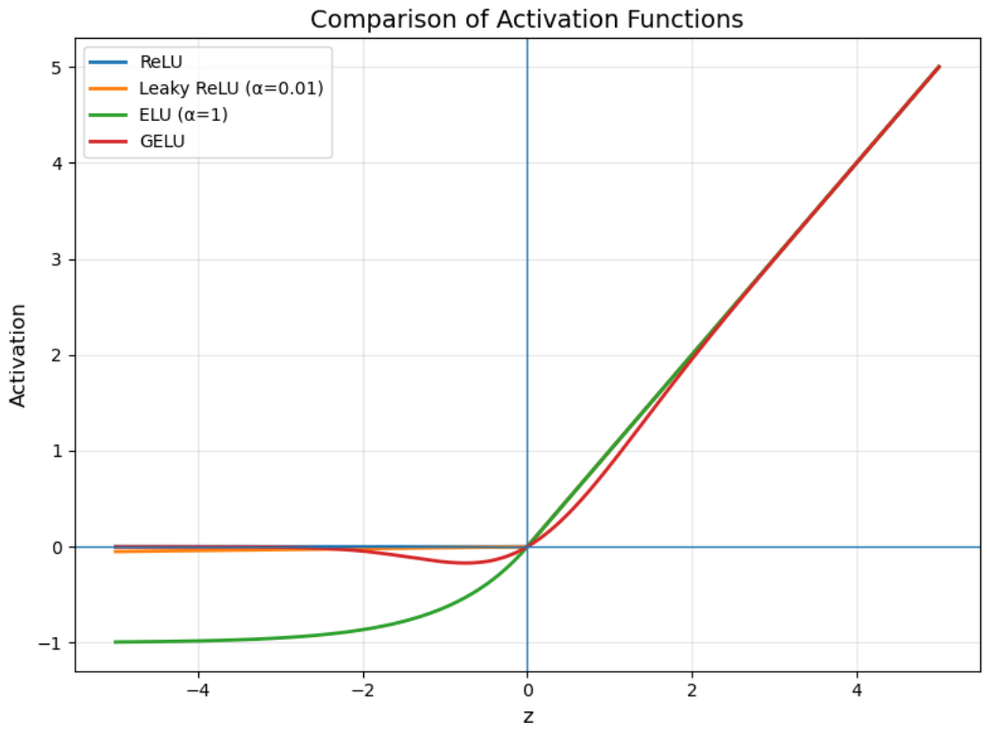
  ELU behaves like a soft biological suppression response rather than hard shut-off.

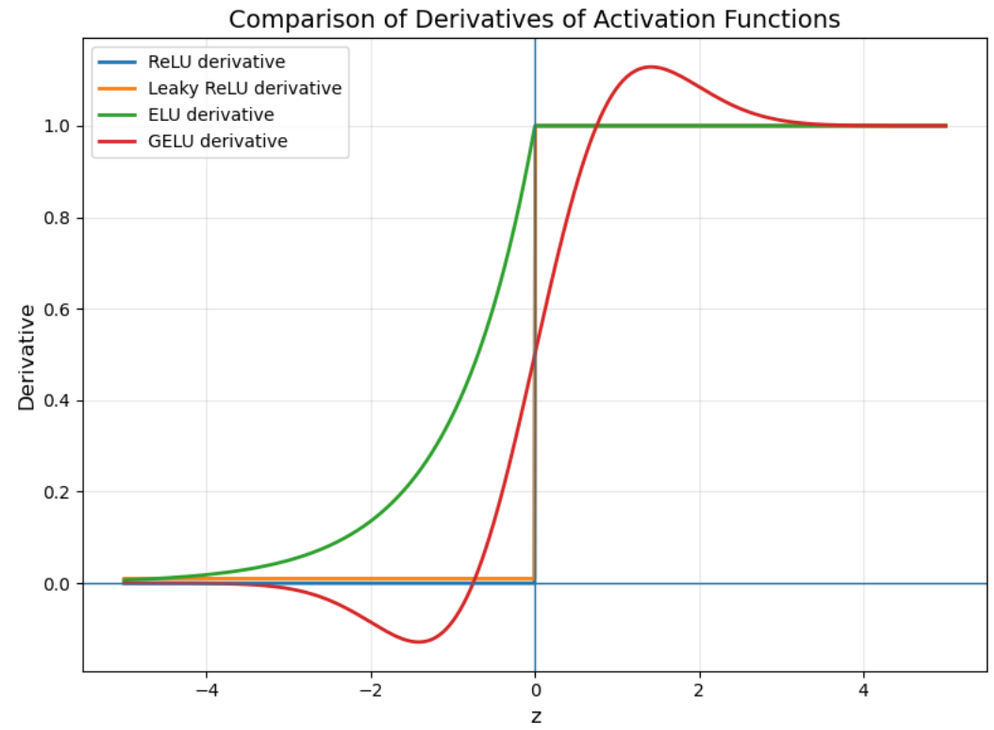
- Gaussian Error Linear Unit (GELU)

$$\text{GELU}(z) = z\Phi(z)$$

  where $\phi(z)$ is the cumulative distribution function of the standard normal distribution. GELU is used in transformers, large language models, and modern deep learning architectures.

Alternative Activation Function: ReLU()





- **ReLU:** derivative is 0 for all $z \leq 0$, which explains how gradients can vanish completely for inactive neurons.
- **Leaky ReLU:** keeps a small nonzero gradient on the negative side, so neurons can still recover.
- **ELU:** smooth derivative on the negative side, approaching 0 gradually.
- **GELU:** smooth, non-piecewise behavior, with a derivative that changes continuously.

Alternative Activation Function: ReLU()

Comparison of ReLU and its variants.

Function	Negative Region	Smoothness	Modern Usage
ReLU	zero flat	non-smooth	very common
ELU	exponential	smooth	moderate
GELU	probabilistic smooth	very smooth	very modern deep learning

Summary:

- Sigmoid: probabilistic intuition
- tanh: optimization intuition
- ReLU: modern deep learning
- ELU: improvement idea
- GELU: frontier concept

Neural Network

What is a Neural Network?

- An artificial neural network is a computational model consisting of interconnected units (neurons) that transform input data into outputs using learnable parameters.
- Neural networks include many architectures:
  - Multilayer perceptrons (MLPs)
  - Convolutional neural networks (CNNs)
  - Recurrent neural networks (RNNs)
  - Transformers
  - Graph neural networks (GNNs)

These architectures differ in how neurons are connected, how information flows, and how inputs are processed.

What is a Neural Network?

Architecture	**Key characteristic**	**Bioinformatics application**
MLP	Fully connected feed-forward layers	Disease prediction from gene expression
CNN	Convolutional filters exploit local patterns	DNA sequence motif recognition
RNN	Recurrent connections process sequential information	Biological time-series modeling
Transformer	Attention mechanisms model relationships among tokens	Protein language models
GNN	Message passing over graph structures	Molecular property prediction

What is a Deep Neural Network?

- A DNN is simply a neural network with multiple hidden layers. It is not a separate architectural family.
- An MLP with one hidden layer is a relatively shallow neural network.
- An MLP with ten hidden layers is both an MLP and a deep neural network.
- A Transformer with many layers is also a deep neural network, but it is not ordinarily described as an MLP as a whole.

Therefore, MLP describes the architecture, whereas deep describes its depth.

Activation Function Usage

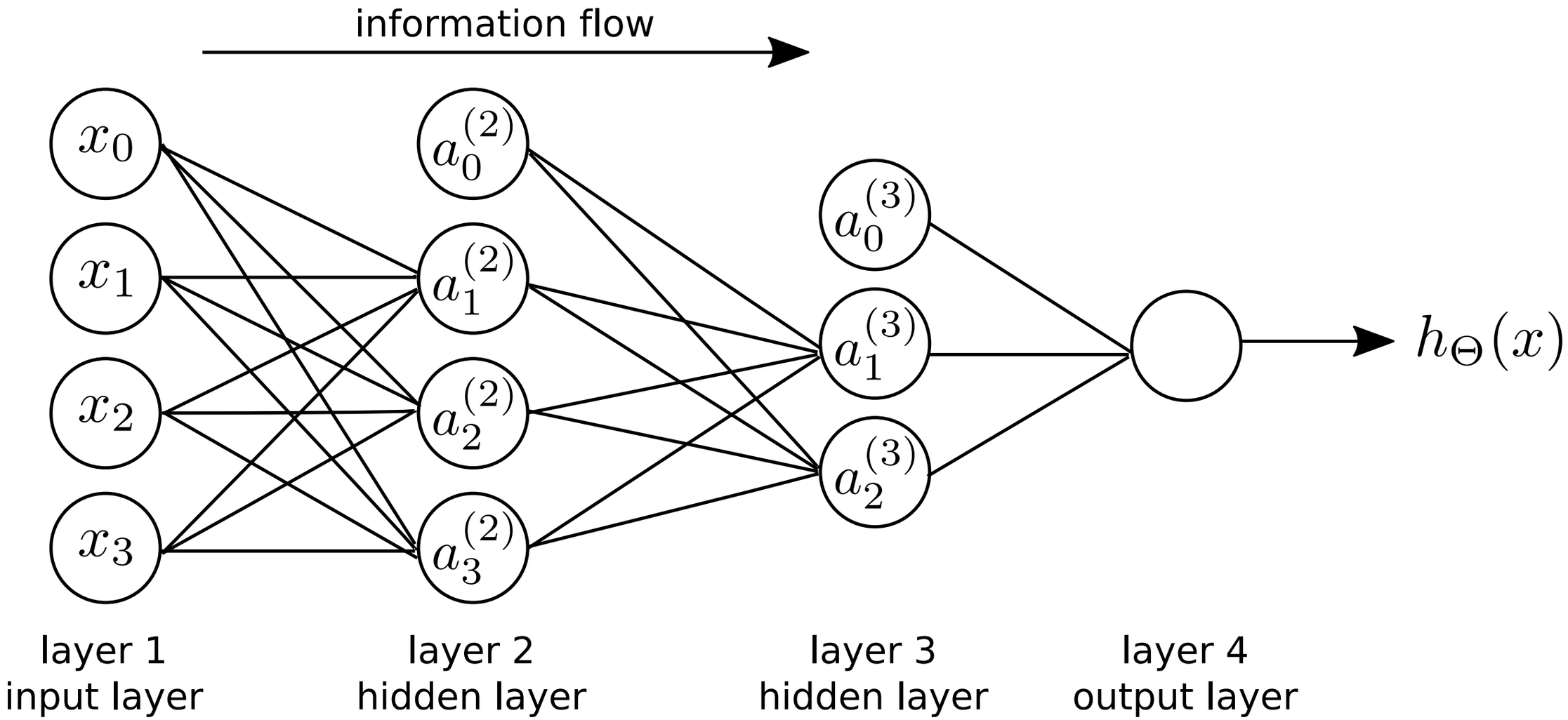
Activation	Typical applications	Comments
ReLU	MLPs, CNNs	Classic default; simple and effective
Leaky ReLU	MLPs, CNNs	Addresses dying ReLU
GELU	Transformers, language models	Smooth activation; widely used in Transformer feed-forward networks
SiLU (Swish)	Modern CNNs, some language models	Smooth activation; used in many newer architectures
Sigmoid	Binary classification output	Less common in hidden layers
Tanh	Traditional RNNs, LSTMs	Less common in feed-forward hidden layers

MLP

What is an MLP?

- An MLP is a feed-forward neural network with one or more hidden layers, typically composed of fully connected (dense) layers.
- An MLP has three defining characteristics:
  - **Feed-forward:** Information moves from input to output without recurrent connections.
  - **Fully connected layers:** Each neuron receives inputs from all neurons in the preceding layer.
  - **Nonlinear activations:** Hidden layers typically apply functions such as ReLU, enabling nonlinear modeling.

General Architecture



Data Transformation

- A neural network does not just fit a function directly.
- It transforms the input data step-by-step into new representations that make prediction easier.

$$x \to h_1 \to h_2 \cdots \to h_\Theta(x)$$

where

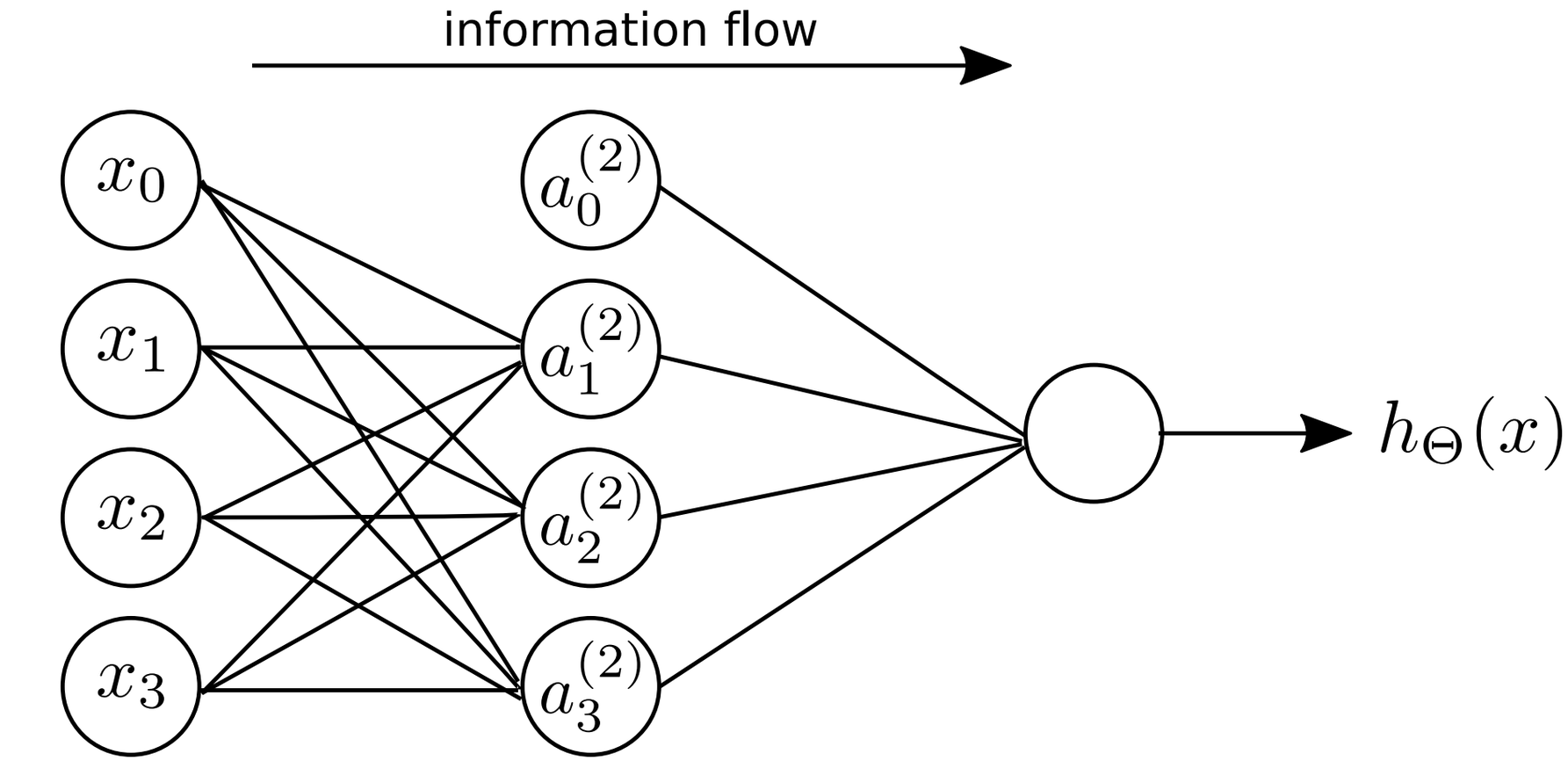
  - x = original biological measurements
  - h_k = hidden layer representation
  - $h_\Theta(x)$ = predicted biological outcome

Hidden layers perform automatic feature engineering.

From One Neuron to Neuron Network

- Logistic regression is a neural network with a single sigmoid output unit and no hidden layers.
- When hidden layers are introduced:
  - The model can learn nonlinear transformations of the input features.
  - Multiple neurons can learn different intermediate representations.
  - The network can model relationships that ordinary logistic regression cannot capture without feature engineering.

ANN Example for Illustration



layer 1
input layer

layer 2
hidden layer

layer 3
output layer

Each neuron is called a unit.

Forward Propagation

- Denote a matrix of weights, Θ, with the Ith row, $\Theta^{(j)}_{I,:}$, representing the function mapping to unit I in layer $j+1$ from units in layer j.
- Denote $a_i^{(j)}$ to represent the "activation" of unit i in layer j.
- Then we have

$$\begin{aligned} a_1^{(2)} &= g\left(\Theta_{10}^{(1)}x_0 + \Theta_{11}^{(1)}x_1 + \Theta_{12}^{(1)}x_2 + \Theta_{13}^{(1)}x_3\right) \\ a_2^{(2)} &= g\left(\Theta_{20}^{(1)}x_0 + \Theta_{21}^{(1)}x_1 + \Theta_{22}^{(1)}x_2 + \Theta_{23}^{(1)}x_3\right) \\ a_3^{(2)} &= g\left(\Theta_{30}^{(1)}x_0 + \Theta_{31}^{(1)}x_1 + \Theta_{32}^{(1)}x_2 + \Theta_{33}^{(1)}x_3\right) \end{aligned}$$

with $\Theta^{(1)} \in \mathbb{R}^{3\times 4}$ and

$$h_\Theta(x) = a_1^{(3)} = g\left(\Theta_{10}^{(2)}a_0^{(2)} + \Theta_{11}^{(2)}a_1^{(2)} + \Theta_{12}^{(2)}a_2^{(2)} + \Theta_{13}^{(2)}a_3^{(2)}\right)$$

with $\Theta^{(2)} \in \mathbb{R}^{1\times 4}$.

Neural Network Representation

- In general, if a network has s_j units in layer j and $s_{(j+1)}$ units in layer $j+1$, then $\Theta^{(j)}$ will be of dimension $s_{(j+1)} \times (s_j + 1)$.

Forward Propagation

- Denote $z_1^{(2)}, z_2^{(2)}, z_3^{(2)}$ as the input to the first, second, and third unit in the hidden layer:

$$\begin{aligned} z_1^{(2)} &= \Theta_{10}^{(1)} x_0 + \Theta_{11}^{(1)} x_1 + \Theta_{12}^{(1)} x_2 + \Theta_{13}^{(1)} x_3 \\ z_2^{(2)} &= \Theta_{20}^{(1)} x_0 + \Theta_{21}^{(1)} x_1 + \Theta_{22}^{(1)} x_2 + \Theta_{23}^{(1)} x_3 \\ z_3^{(2)} &= \Theta_{30}^{(1)} x_0 + \Theta_{31}^{(1)} x_1 + \Theta_{32}^{(1)} x_2 + \Theta_{33}^{(1)} x_3 \end{aligned}$$

and denote

$$z^{(2)} = \begin{bmatrix} z_1^{(2)} \\ z_2^{(2)} \\ z_3^{(2)} \end{bmatrix}$$

- Then we have

$$\begin{aligned} z^{(2)} &= \Theta^{(1)} x \\ a^{(2)} &= g\left(z^{(2)}\right) \end{aligned}$$

where $g(\cdot)$ applies to each element of $z^{(2)}$ separately.

Forward Propagation

- Define

$$a^{(1)} := x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

then

$$\begin{aligned} z^{(2)} &= \Theta^{(1)} a^{(1)} \\ a^{(2)} &= g\left(z^{(2)}\right) \end{aligned}$$

with $a^{(1)} \in \mathbb{R}^{4\times 1}$, $\Theta^{(1)} \in \mathbb{R}^{3\times 4}$, $z^{(2)} \in \mathbb{R}^{3\times 1}$ and $a^{(2)} \in \mathbb{R}^{3\times 1}$.

- Add $a_0^{(2)} = 1$, then we have

$$\begin{aligned} z^{(3)} &= \Theta^{(2)} a^{(2)} \\ h_\Theta(x) &= a^{(3)} = g\left(z^{(3)}\right) \end{aligned}$$

Forward Propagation

- In general,

$$\begin{aligned} z^{(j)} &= \Theta^{(j-1)} a^{(j-1)} \\ a^{(j)} &= g\left(z^{(j)}\right) \end{aligned}$$

- If the network has a total of n layers, then

$$h_\Theta(x) = a^{(n)} = g\left(z^{(n)}\right)$$

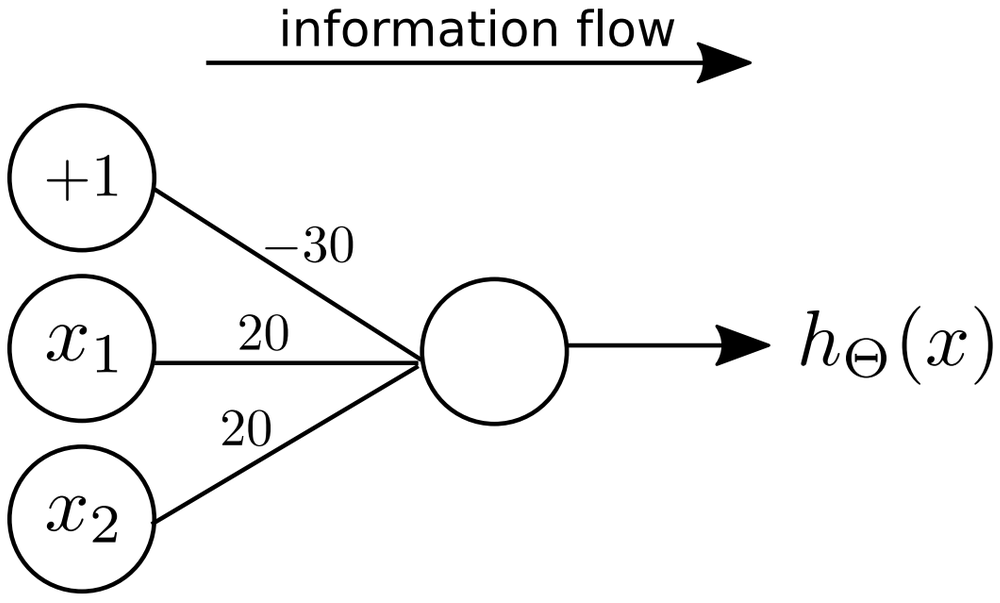
- Remark: The forward propagation in a neural network consists of a series of logistic regression.

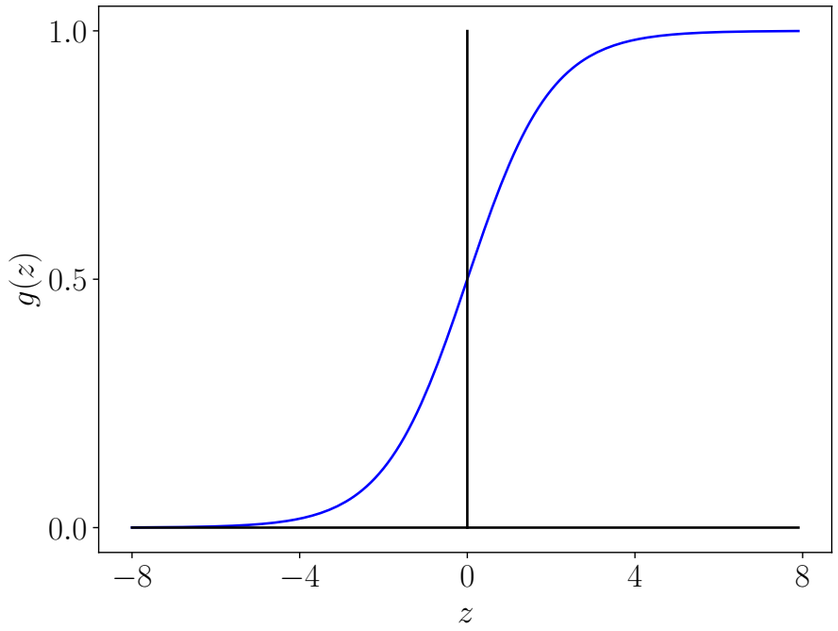
Forward Propagation

When the bias term is explicit, the general forward propagation is described as:

$$z^{(l)} = W^{(l)} a^{(l-1)} + b^{(l)}$$
$$a^{(l)} = g^{(l)}(z^{(l)})$$

Example: logical AND function

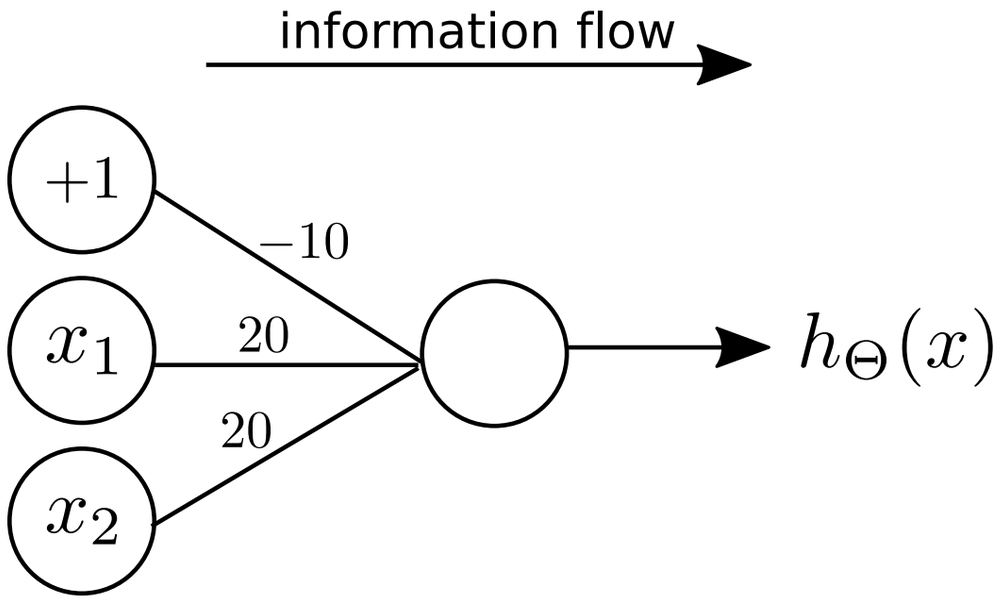




- $x_1, x_2 \in \{0, 1\}$
- $y = x_1 \text{ AND } x_2$
- $h_\Theta(x) = -30 + 20x_1 + 20x_2$

x_1	x_2	$h_\Theta(x)$
0	0	$g(-30) \approx 0$
0	1	$g(-10) \approx 0$
1	0	$g(-10) \approx 0$
1	1	$g(+10) \approx 1$

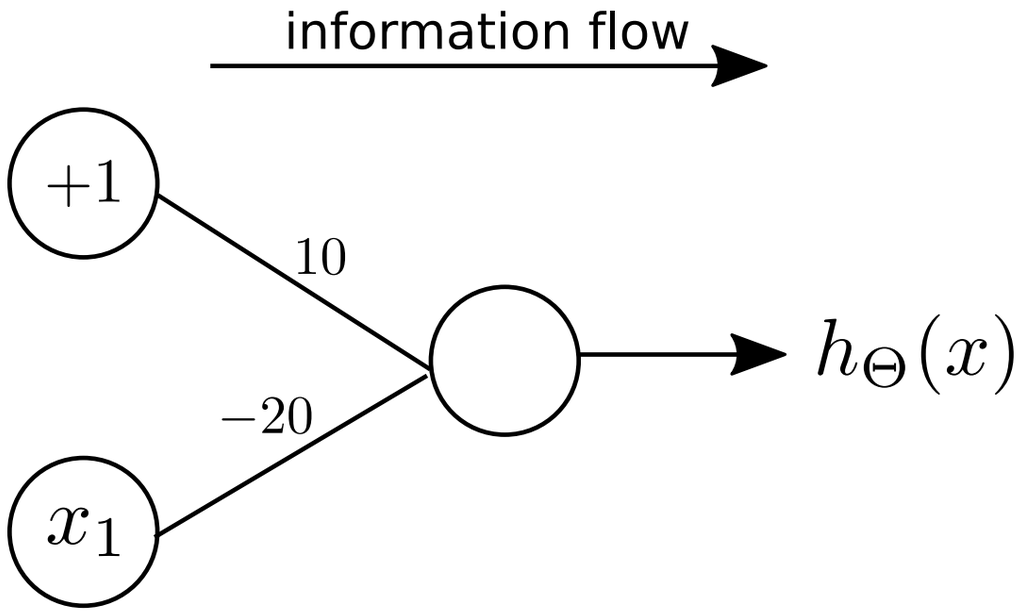
Example: logical OR function



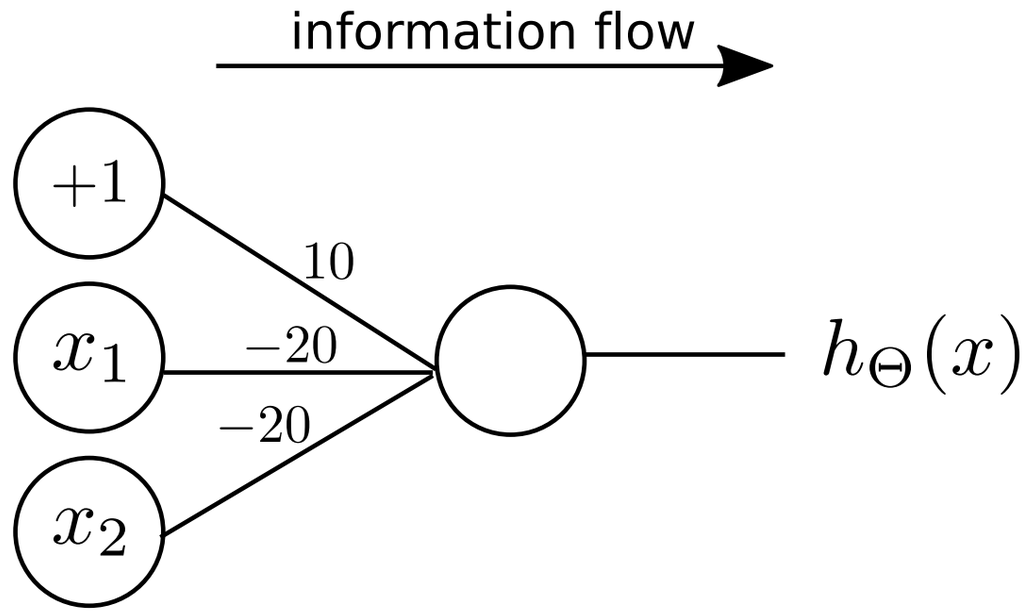
- $x_1, x_2 \in \{0, 1\}$
- $y = x_1 \text{ OR } x_2$
- $h_\Theta(x) = -10 + 20x_1 + 20x_2$

Example: Other logical functions

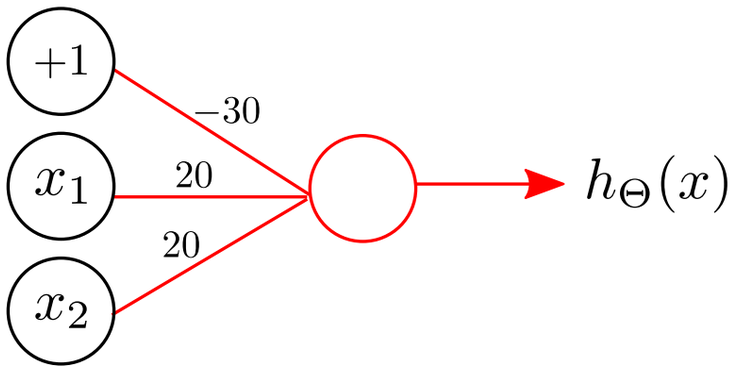
NOT x_1



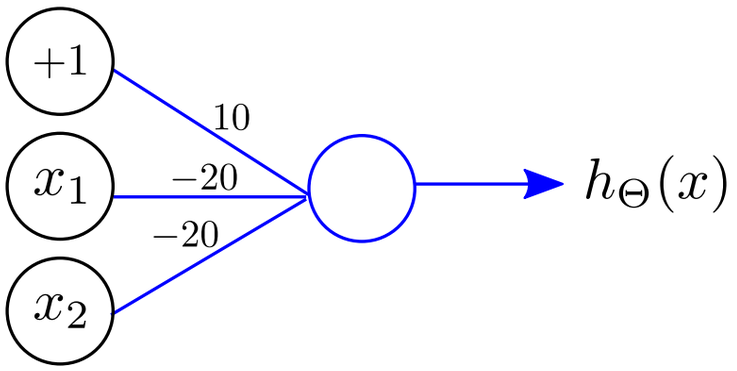
(NOT x_1) AND (NOT x_2)



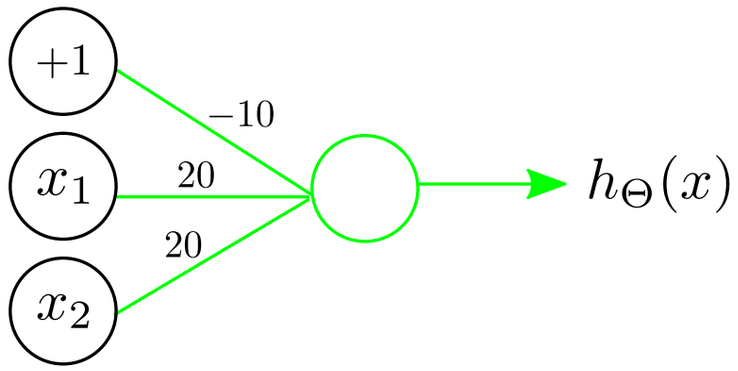
Example: logical XNOR



x_1 AND x_2

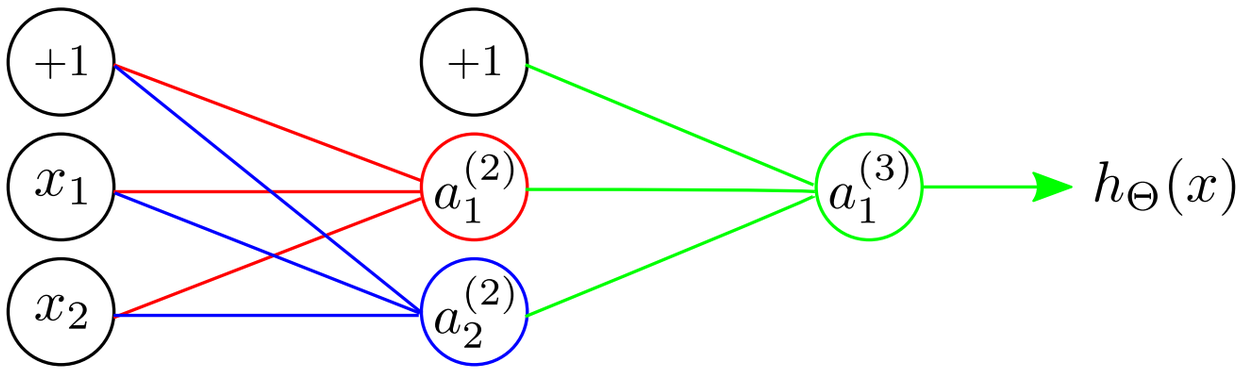


(NOT x_1) AND (NOT x_2)



x_1 OR x_2

Putting together:



x_1	x_2	$a_1^{(2)}$	$a_2^{(2)}$	$h_\Theta(x)$
0	0	0	1	1
0	1	0	0	0
1	0	0	0	0
1	1	1	0	1

with

$$\Theta^{(1)} = \begin{bmatrix} -30 & 20 & 20 \\ 10 & -20 & -20 \end{bmatrix}, \quad \Theta^{(2)} = \begin{bmatrix} -10 & 20 & 20 \end{bmatrix}$$

and

$$a^{(2)} = g\left(\Theta^{(1)} \cdot x\right), \quad a^{(3)} = g\left(\Theta^{(2)} \cdot a^{(2)}\right), \quad h_\Theta(x) = a^{(3)}$$

Neural Network Learning

Cost Function In General

Mean Squared Error (MSE)

- Definition

$$J = \frac{1}{2m} \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)}) \right)^2$$

- Used for regression tasks where continuous phenotype is predicted, e.g., concentration of cholesterol.

Binary cross-entropy loss

- Definition

$$J = -\frac{1}{2m} \left[y \log(h_\theta(x^{(i)}) + (1-y) \log(1 - h_\theta(x^{(i)}) \right]$$

Overall Training Cycle

- Step 1: Forward Propagation

  Input data moves through network:

  $$x \to h_1 \to h_2 \to \cdots \to h_\Theta(x)$$

  and produces prediction.

- Step 2: Compute Cost

- Step 3: Back Propagation

  We compute:

  $$\frac{\partial J}{\partial \theta}$$

  for every θ in the network. This tells us which weights contributed to the cost and by how much.

- Step 4: Parameter Update

  Gradient descent update:

  $$\theta \leftarrow \theta - \alpha \frac{\partial J}{\partial \theta}$$

  Learning improves prediction.

Neural Network for Classification

- Notations

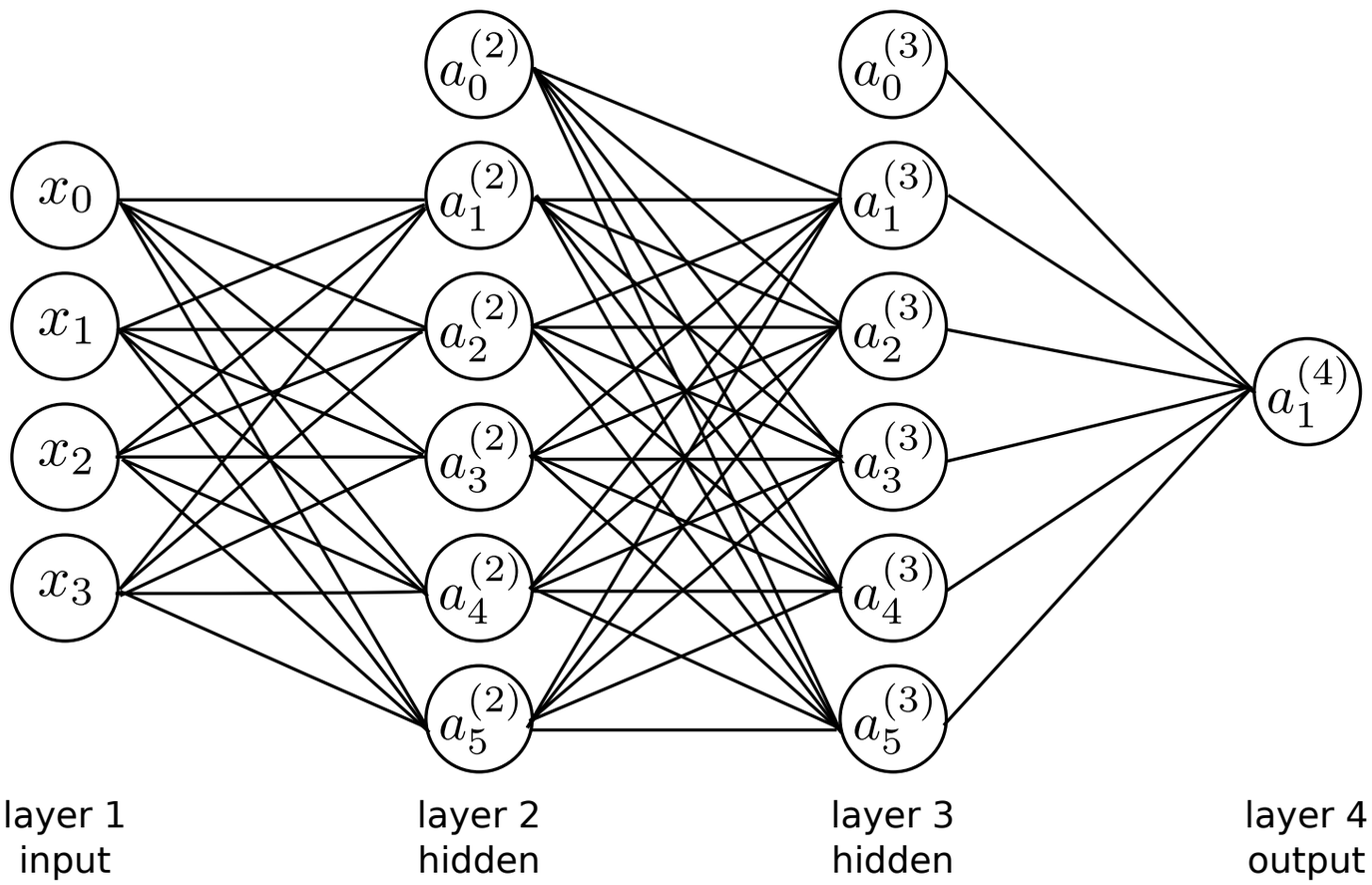
  Samples: $\{\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \cdots, \left(x^{(m)}, y^{(m)}\right)\}$

  L: total number of layers in the network

  s_l: number of units (not counting bias unit) in layer l

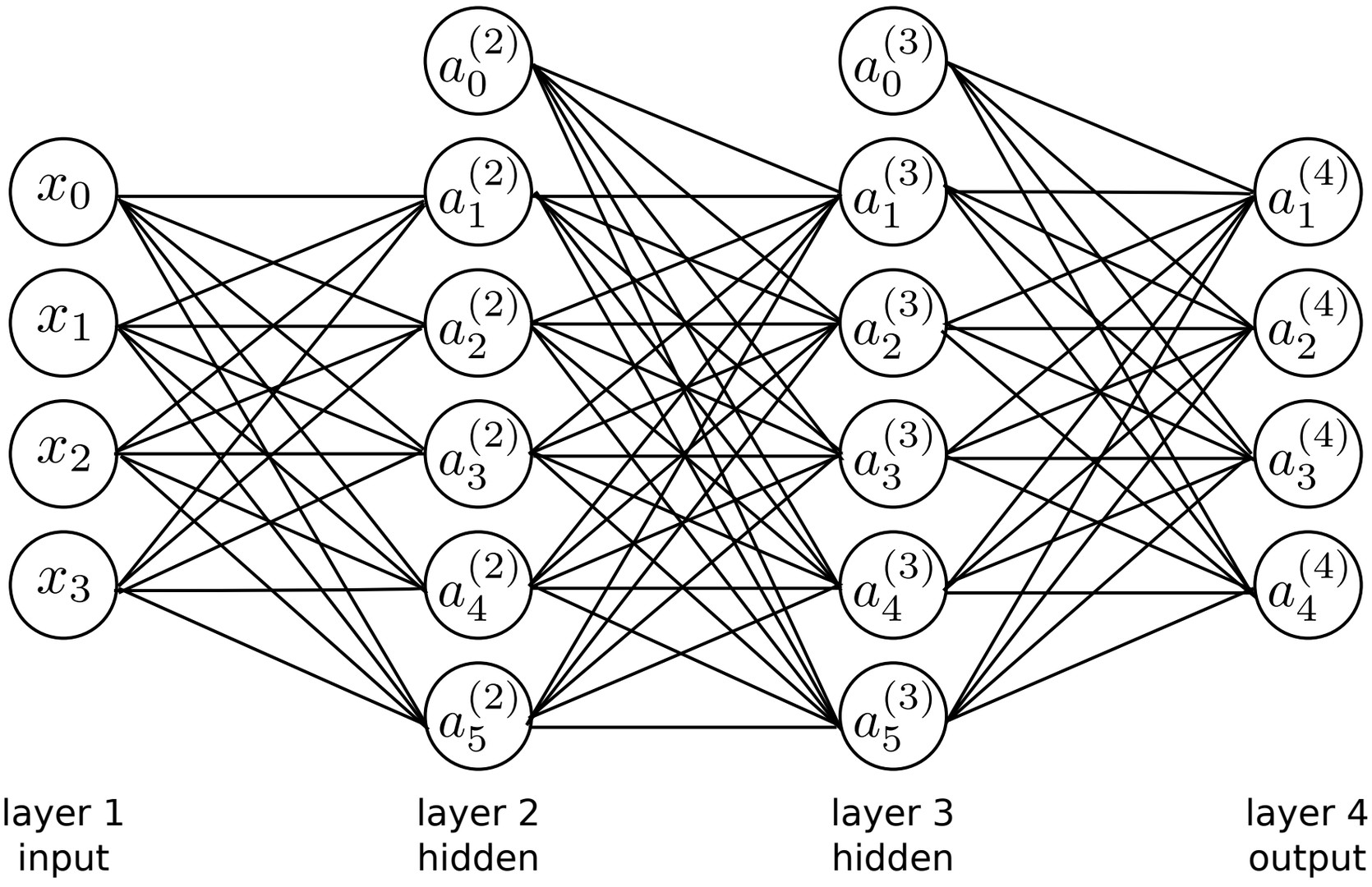
  K: number of classes

Binary Classification



$$y \in \{0, 1\}$$
$$K = 1,\ h_\Theta(x) \in \mathbb{R}^K,\ \text{i.e. } h_\Theta(x) \in \mathbb{R}$$
$$L = 4,\ s_1 = 3, s_2 = 5, s_3 = 5, s_4 = 1$$

Multiclass Classification



Multiclass Classification

- This is for $K \geq 3$ classification.
- For the example on the previous slide, $K = 4$, $y \in \mathbb{R}^K$ and we use

$$y = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

to represent 4 classes, e.g., pedestrian, car, motorcycle, and truck, respectively.

$L = 4,\ s_1 = 3, s_2 = 5, s_3 = 5, s_4 = 4$

$h_\Theta(x) \in \mathbb{R}^4$

Cost Function for Regression

- Use the mean squared error
- Linear regression

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (\hat{y}_i - y_i)^2 = \frac{1}{2m} \sum_{i=1}^{m} \left(\theta^T x_i - y_i\right)^2$$

- Neural network

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \sum_{k=1}^{K} \left[\left(h_\theta(x^{(i)}) \right)_k - y_k^{(i)} \right]^2$$

where K is the total number of output units.

Minimize the Cost Function

- Next we need to find Θ that would minimize $J(\Theta)$.
- The minimization requires code to compute
  - $J(\Theta)$
  - $\dfrac{\partial}{\partial \Theta_{ij}^{(l)}} J(\Theta)$

- Computation of $J(\Theta)$ is straightforward.
- The optimal Θ can be found by gradient descent.

Gradient Descent Analogy

Imagine:

- Cost function = mountain landscape
- parameter values determine your current location
- High cost → high altitude and low cost → low altitude

Training process: gradient descent is like walking downhill trying to reach the lowest valley.

Gradient and Gradient Descent

- Gradient is defined as:

$$\nabla J(\theta) = \begin{bmatrix} \frac{\partial J}{\partial \theta_1} \\ \frac{\partial J}{\partial \theta_1} \\ \vdots \\ \frac{\partial J}{\partial \theta_d} \end{bmatrix}$$

- Gradient tells us which direction increases J fastest.
- So negative gradient tells us the direction of steepest decrease in J.
- Gradient descent tells us how to update the θ values based on the current values as follows:

$$\theta_{\text{new}} := \theta_{\text{old}} - \alpha \nabla J(\theta)$$

where α, a positive real number, is the learning rate.

Training Cycle

- ANN makes prediction using forward propagation.
- Compute cost.
- Compute gradient using back propagation.
- Update parameters Θ.

Repeat this process thousands of times.

Gradient Descent Example

- Cost function:

$$J(\theta) = (\theta - 3)^2$$

- Find θ where J is the minimum.
- Gradient:

$$\frac{dJ}{d\theta} = 2(\theta - 3)$$

- θ update rule:

$$\theta_{\text{new}} = \theta_{\text{old}} - \alpha \cdot 2(\theta - 3)$$

- We can see that:
  - $\theta < 3 \rightarrow$ gradient is negative $\rightarrow \theta_{\text{new}}$ moves right
  - $\theta < 3 \rightarrow$ gradient is positive $\rightarrow \theta_{\text{new}}$ moves left

Learning Rate Intuition

Learning rate α intuition

- Learning rate controls the step size when walking downhill.
- Too small $\rightarrow$ very slow training
- Too large $\rightarrow$ overshoot valley and unstable training
- Good learning rate $\rightarrow$ smooth convergence.

Analogy: like adjusting walking stride when descending a mountain.

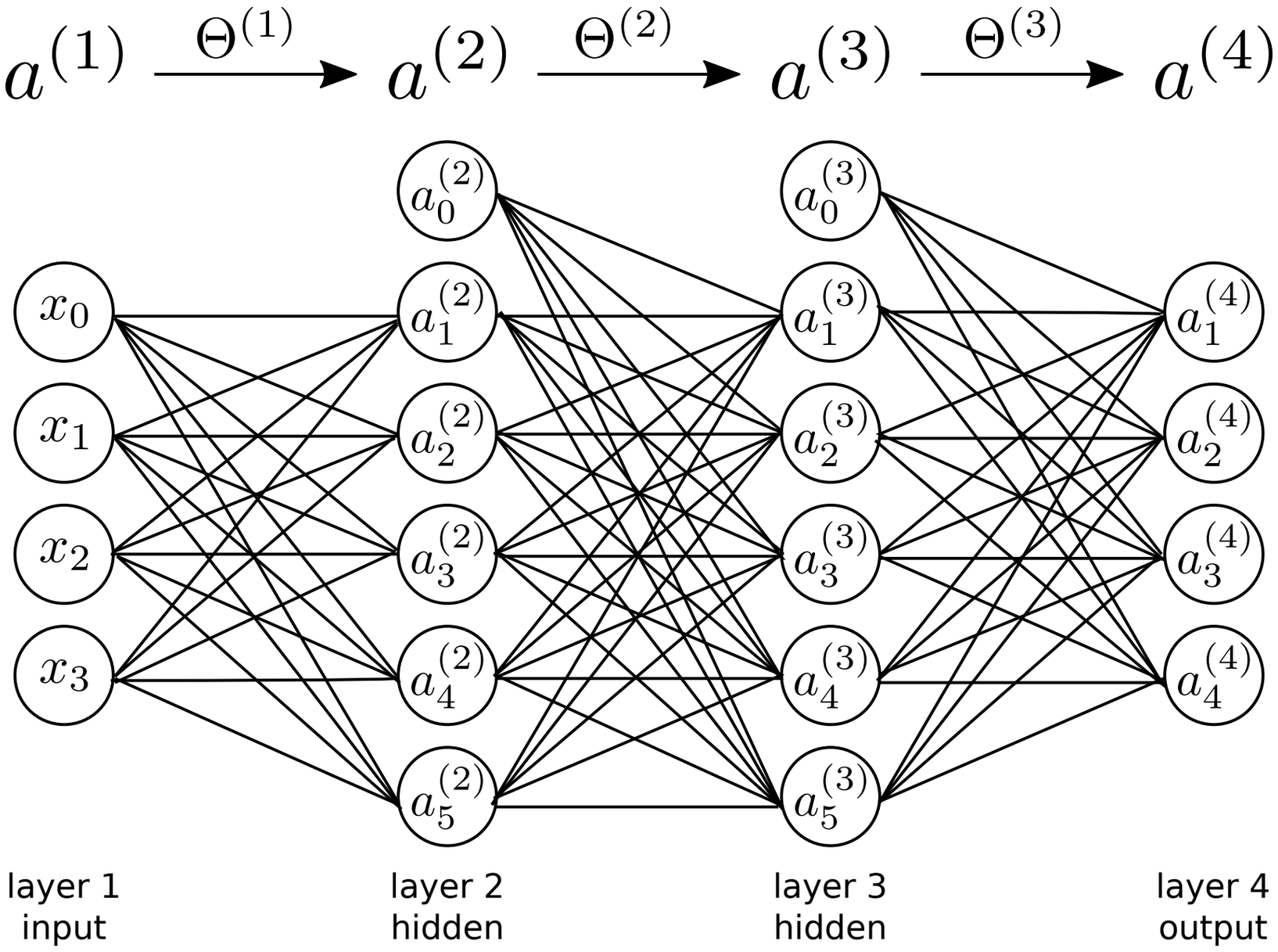
High-Dimensional Reality in ANN

- Real ANN has millions of parameters and the cost function surface is extremely complex.
- Gradient descent guides the navigation in a high-dimensional error landscape to reach the minimum.
- Training is an iterative improvement and there is no instant solution.

Local vs Global Minimum

- Neural network creates a cost function surface that is:
  - extremely high-dimensional
  - highly non-convex
  - full of complex curvature
- This cost function surface is characterized by:
  - many valleys
  - flat plateaus
  - saddle points
- So the optimization becomes a very complex navigation problem.
- Usually the final minimum reached during neural network training is a local minimum (or saddle-region), not guaranteed to be the global minimum.
- In deep learning, many local minima have similar cost values and some local minima generalize better. Therefore, reaching global minimum is not always necessary.

Summary: Forward Propagation



Summary: Forward Propagation

- Given one training sample (x, y), forward propagation is done as follows for the neural network architecture shown on the previous slide:

$$
\begin{aligned}
a^{(1)} &= x \\
z^{(2)} &= \Theta^{(1)} a^{(1)} \\
a^{(2)} &= g\left(z^{(2)}\right) \\
&\quad \text{then add } a_0^{(2)} \text{ to } a^{(2)} \\
z^{(3)} &= \Theta^{(2)} a^{(2)} \\
a^{(3)} &= g\left(z^{(3)}\right) \\
&\quad \text{then add } a_0^{(3)} \text{to } a^{(3)} \\
z^{(4)} &= \Theta^{(3)} a^{(3)} \\
a^{(4)} &= g\left(z^{(4)}\right) = h_\Theta(x)
\end{aligned}
$$

Back Propagation

- Neural networks contain:
  - many layers
  - millions of parameters
- We cannot adjust weights by intuition. Back propapation provides **systematic credit assignment mechanism**
- Back propagation relies on the **chain rule of calculus**

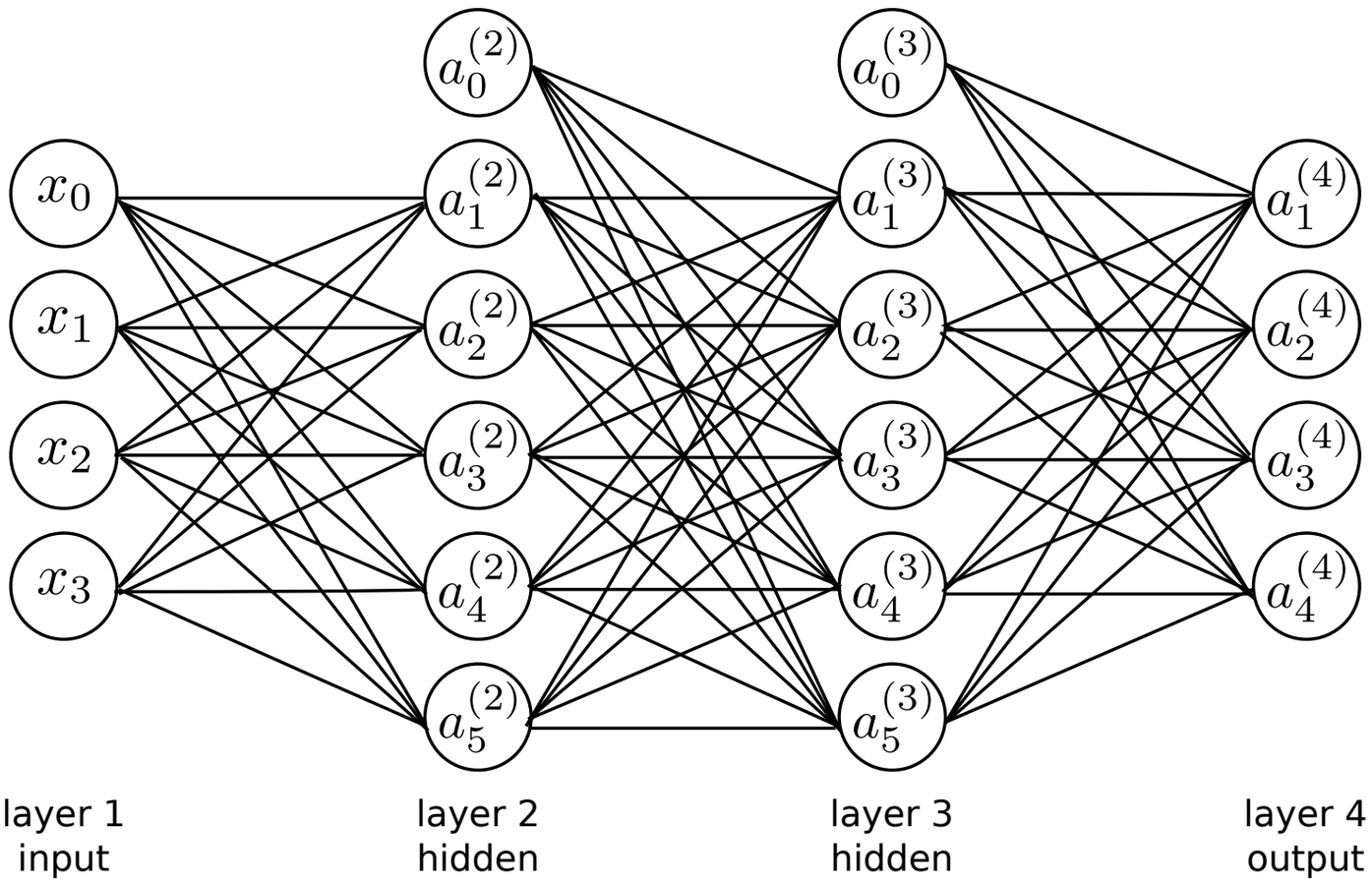
$$\theta \to z \to h_\Theta \to J$$

Then:

$$\frac{\partial J}{\partial \theta} = \frac{\partial J}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial \theta}$$

Back Propagation

$$\delta^{(2)} \xleftarrow[a^{(2)}]{\Theta^{(2)}} \delta^{(3)} \xleftarrow[a^{(3)}]{\Theta^{(3)}} \delta^{(4)}$$

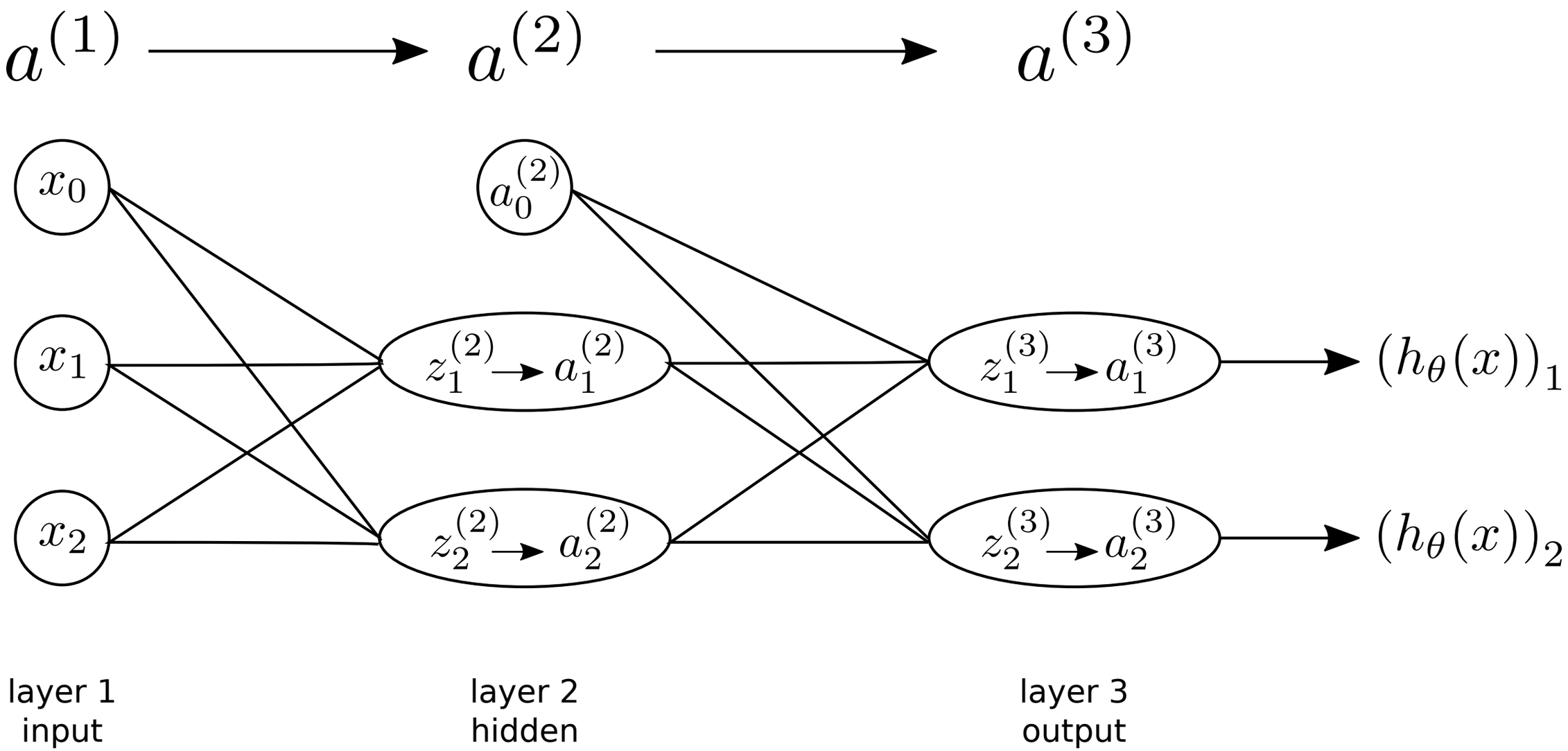


- Use back propagation to calculate

$$\frac{\partial}{\partial \Theta_{ij}^{(l)}} J(\Theta)$$

for all i, j and $l = 1, 2, \cdots, L-1$.

Example: Forward Propagation



Example: Forward Propagation

- From $a^{(1)}$ to $a^{(2)}$:

$$z^{(2)} = \begin{bmatrix} z_1^{(2)} \\ \\ z_2^{(2)} \end{bmatrix} = \begin{bmatrix} \Theta_{10}^{(1)} & \Theta_{11}^{(1)} & \Theta_{12}^{(1)} \\ \\ \Theta_{20}^{(1)} & \Theta_{21}^{(1)} & \Theta_{22}^{(1)} \end{bmatrix} \begin{bmatrix} a_0^{(1)} \\ \\ a_1^{(1)} \\ \\ a_2^{(1)} \end{bmatrix}, \quad a^{(2)} = \begin{bmatrix} a_1^{(2)} \\ \\ a_2^{(2)} \end{bmatrix} = g\left(z^{(2)}\right)$$

- From $a^{(2)}$ to $a^{(3)}$:

$$z^{(3)} = \begin{bmatrix} z_1^{(3)} \\ \\ z_2^{(3)} \end{bmatrix} = \begin{bmatrix} \Theta_{10}^{(2)} & \Theta_{11}^{(2)} & \Theta_{12}^{(2)} \\ \\ \Theta_{20}^{(2)} & \Theta_{21}^{(2)} & \Theta_{22}^{(2)} \end{bmatrix} \begin{bmatrix} a_0^{(2)} \\ \\ a_1^{(2)} \\ \\ a_2^{(2)} \end{bmatrix}, \quad a^{(3)} = \begin{bmatrix} a_1^{(3)} \\ \\ a_2^{(3)} \end{bmatrix} = g\left(z^{(3)}\right)$$

- Total cost:

$$J(\Theta) = \frac{1}{2}\left[\left(y_1 - a_1^{(3)}\right)^2 + \left(y_2 - a_2^{(3)}\right)^2\right]$$

Example: Back Propagation

- The goal is to calculate $\frac{\partial J\Theta}{\partial \Theta_{ij}^{(l)}}$ for $l = 1, 2$. We will first compute these partial derivatives for $l = 2$ and then for $l = 1$.
- $l = 2$:

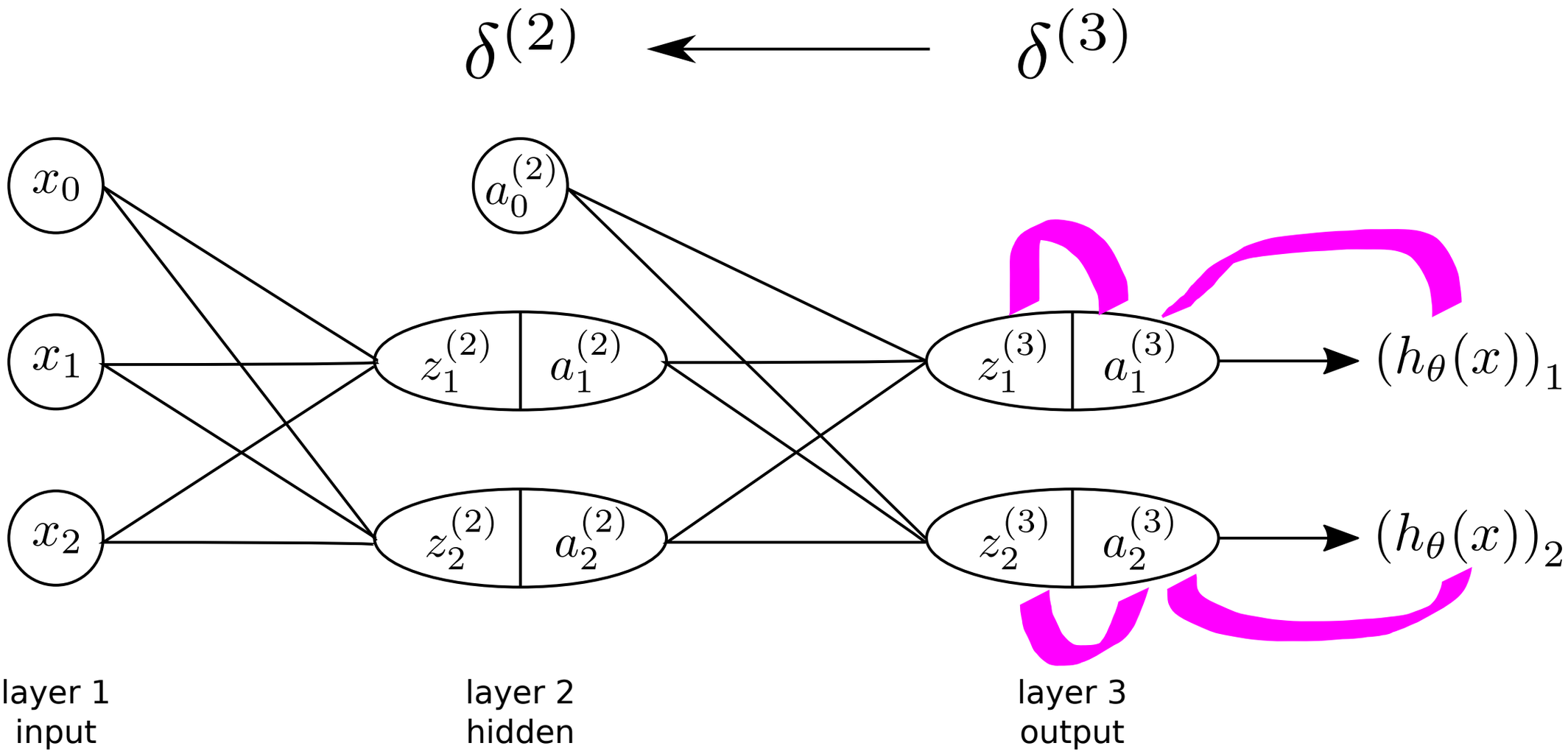
$$\left[\frac{\partial J(\Theta)}{\partial \Theta_{10}^{(2)}}, \frac{\partial J(\Theta)}{\partial \Theta_{11}^{(2)}}, \frac{\partial J(\Theta)}{\partial \Theta_{12}^{(2)}}\right] = \frac{\partial J(\Theta)}{\partial z_1^{(3)}} \cdot \left[a_0^{(2)}, a_1^{(2)}, a_2^{(2)}\right]$$

and

$$\left[\frac{\partial J(\Theta)}{\partial \Theta_{20}^{(2)}}, \frac{\partial J(\Theta)}{\partial \Theta_{21}^{(2)}}, \frac{\partial J(\Theta)}{\partial \Theta_{22}^{(2)}}\right] = \frac{\partial J(\Theta)}{\partial z_2^{(3)}} \cdot \left[a_0^{(2)}, a_1^{(2)}, a_2^{(2)}\right]$$

where $\frac{\partial J(\Theta)}{\partial z_1^{(3)}} \in \mathbb{R}$ and $\frac{\partial J(\Theta)}{\partial z_2^{(3)}} \in \mathbb{R}$.

Example: Back Propagation



Example: Back Propagation

- By the chain rule in calculus, we have

$$\frac{\partial J(\Theta)}{\partial z_1^{(3)}} = \frac{\partial J(\Theta)}{\partial a_1^{(3)}} \cdot \frac{\partial a_1^{(3)}}{\partial z_1^{(3)}} = \left(a_1^{(3)} - y_1\right) \cdot \left[a_1^{(3)} \cdot \left(1 - a_1^{(3)}\right)\right]$$

  with

$$\frac{\partial J(\Theta)}{\partial a_1^{(3)}} = a_1^{(3)} - y_1$$

  for the mean squared error cost function and

$$\frac{\partial a_1^{(3)}}{\partial z_1^{(3)}} = a_1^{(3)} \cdot \left(1 - a_1^{(3)}\right)$$

  for the logistic activation function.

- Similarly,

$$\frac{\partial J(\Theta)}{\partial z_2^{(3)}} = \frac{\partial J(\Theta)}{\partial a_2^{(3)}} \cdot \frac{\partial a_2^{(3)}}{\partial z_2^{(3)}} = \left(a_2^{(3)} - y_2\right) \cdot \left[a_2^{(3)} \cdot \left(1 - a_2^{(3)}\right)\right]$$

Example: Back Propagation

- Define

$$\delta^{(3)} := \begin{bmatrix} \dfrac{\partial J(\Theta)}{\partial z_1^{(3)}} \\ \\ \dfrac{\partial J(\Theta)}{\partial z_2^{(3)}} \end{bmatrix} = \begin{bmatrix} \left(a_1^{(3)} - y_1\right) \cdot \left[a_1^{(3)} \cdot \left(1 - a_1^{(3)}\right)\right] \\ \\ \left(a_2^{(3)} - y_2\right) \cdot \left[a_2^{(3)} \cdot \left(1 - a_2^{(3)}\right)\right] \end{bmatrix}$$

- Then

$$\begin{bmatrix} \dfrac{\partial J(\Theta)}{\partial \Theta_{10}^{(2)}}, \dfrac{\partial J(\Theta)}{\partial \Theta_{11}^{(2)}}, \dfrac{\partial J(\Theta)}{\partial \Theta_{12}^{(2)}} \\ \\ \dfrac{\partial J(\Theta)}{\partial \Theta_{20}^{(2)}}, \dfrac{\partial J(\Theta)}{\partial \Theta_{21}^{(2)}}, \dfrac{\partial J(\Theta)}{\partial \Theta_{22}^{(2)}} \end{bmatrix} = \delta^{(3)} \cdot \left[a_0^{(2)}, a_1^{(2)}, a_2^{(2)}\right]$$

Example: Back Propagation

- Calculate $\dfrac{\partial J\Theta}{\partial \Theta_{ij}^{(l)}}$ for $l = 1$:

$$\left[\frac{\partial J(\Theta)}{\partial \Theta_{10}^{(1)}}, \frac{\partial J(\Theta)}{\partial \Theta_{11}^{(1)}}, \frac{\partial J(\Theta)}{\partial \Theta_{12}^{(1)}}\right] = \frac{\partial J(\Theta)}{\partial z_1^{(2)}} \cdot \left[a_0^{(1)}, a_1^{(1)}, a_2^{(1)}\right]$$

and

$$\left[\frac{\partial J(\Theta)}{\partial \Theta_{20}^{(1)}}, \frac{\partial J(\Theta)}{\partial \Theta_{21}^{(1)}}, \frac{\partial J(\Theta)}{\partial \Theta_{22}^{(1)}}\right] = \frac{\partial J(\Theta)}{\partial z_2^{(2)}} \cdot \left[a_0^{(1)}, a_1^{(1)}, a_2^{(1)}\right]$$

where $\dfrac{\partial J(\Theta)}{\partial z_1^{(2)}} \in \mathbb{R}$ and $\dfrac{\partial J(\Theta)}{\partial z_2^{(2)}} \in \mathbb{R}$.

Example: Back Propagation

- The total cost is the sum of the costs for unit 1 and unit 2 in the output layer:

$$J(\Theta) = \frac{1}{2m} \sum_{i=1}^{m} \sum_{k=1}^{K} \left[\left(h_\Theta(x^{(i)}) \right)_k - y_k^{(i)} \right]^2$$

- For a single sample $\{(x, y)\}$, the total cost can be written as

$$J(\Theta) = \frac{1}{2} \left[\left(a_1^{(3)} - y_1 \right)^2 + \left(a_2^{(3)} - y_2 \right)^2 \right]$$

- Define $J_1(\Theta)$ and $J_2(\Theta)$ as the cost for unit 1 and unit 2 in the output layer, respectively, i.e.

$$J_1(\Theta) = \frac{1}{2} \left(a_1^{(3)} - y_1 \right)^2, \quad J_2(\Theta) = \frac{1}{2} \left(a_2^{(3)} - y_2 \right)^2$$

- Then

$$J(\Theta) = J_1(\Theta) + J_2(\Theta)$$

Example: Back Propagation

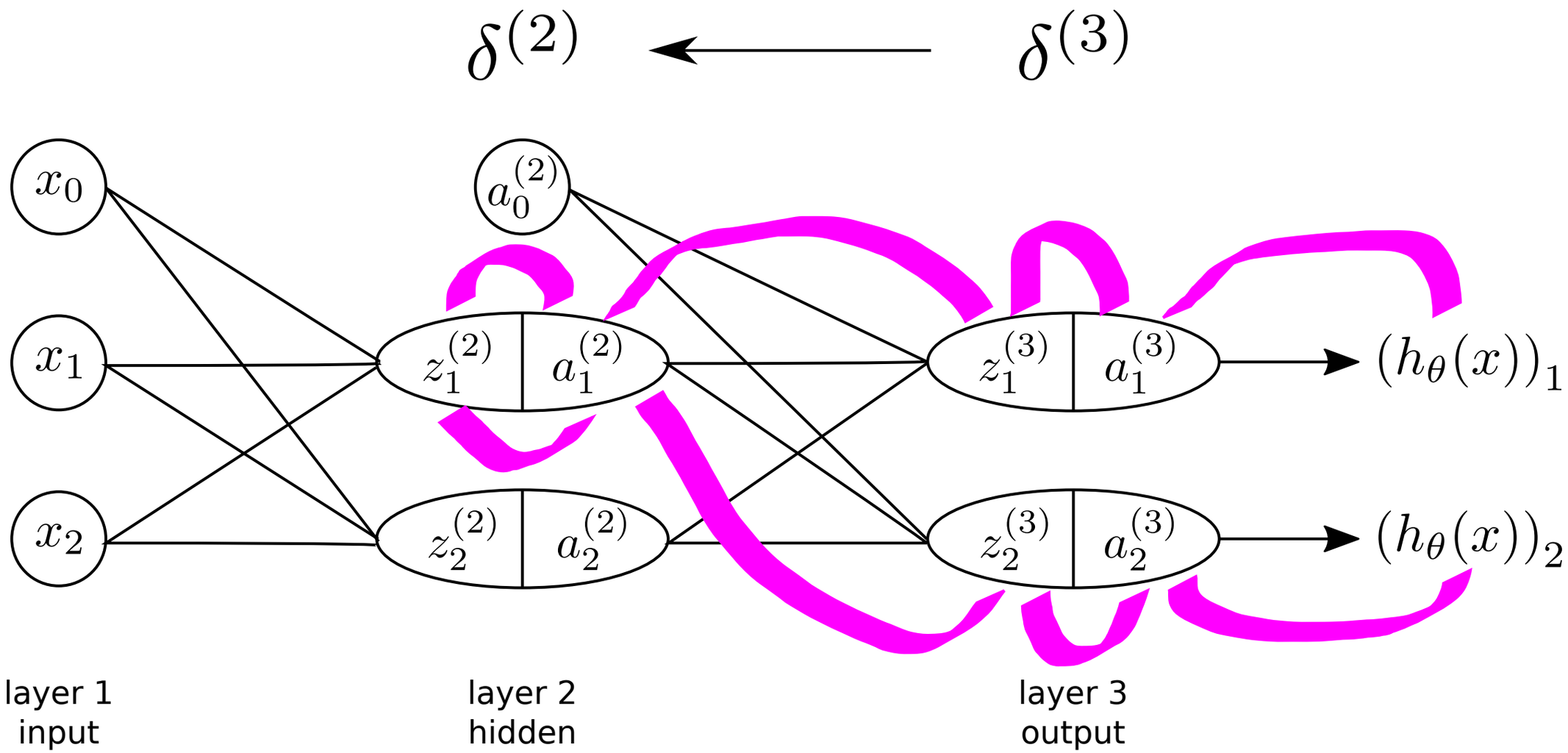
- Now we compute $\dfrac{\partial J(\Theta)}{\partial z_1^{(2)}}$.

$$\frac{\partial J(\Theta)}{\partial z_1^{(2)}} = \frac{\partial J_1(\Theta)}{\partial z_1^{(2)}} + \frac{\partial J_2(\Theta)}{\partial z_1^{(2)}}$$

with

$$\begin{aligned}
\frac{\partial J_1(\Theta)}{\partial z_1^{(2)}} &= \frac{\partial J_1(\Theta)}{\partial a_1^{(3)}} \cdot \frac{\partial a_1^{(3)}}{\partial z_1^{(3)}} \cdot \frac{\partial z_1^{(3)}}{\partial a_1^{(2)}} \cdot \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \\
&= \left(a_1^{(3)} - y_1\right) \cdot \left[a_1^{(3)}\left(1 - a_1^{(3)}\right)\right] \cdot \Theta_{11}^{(2)} \cdot \left[a_1^{(2)}\left(1 - a_1^{(2)}\right)\right] \\
\frac{\partial J_2(\Theta)}{\partial z_1^{(2)}} &= \frac{\partial J_2(\Theta)}{\partial a_2^{(3)}} \cdot \frac{\partial a_2^{(3)}}{\partial z_2^{(3)}} \cdot \frac{\partial z_2^{(3)}}{\partial a_1^{(2)}} \cdot \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \\
&= \left(a_2^{(3)} - y_2\right) \cdot \left[a_2^{(3)}\left(1 - a_2^{(3)}\right)\right] \cdot \Theta_{21}^{(2)} \cdot \left[a_1^{(2)}\left(1 - a_1^{(2)}\right)\right]
\end{aligned}$$

Example: Back Propagation



Example: Back Propagation

- Similarly for $\dfrac{\partial J(\Theta)}{\partial z_2^{(2)}}$, we have

$$\frac{\partial J(\Theta)}{\partial z_2^{(2)}} = \frac{\partial J_1(\Theta)}{\partial z_2^{(2)}} + \frac{\partial J_2(\Theta)}{\partial z_2^{(2)}}$$

with

$$\begin{aligned}
\frac{\partial J_1(\Theta)}{\partial z_2^{(2)}} &= \frac{\partial J_1(\Theta)}{\partial a_1^{(3)}} \cdot \frac{\partial a_1^{(3)}}{\partial z_1^{(3)}} \cdot \frac{\partial z_1^{(3)}}{\partial a_2^{(2)}} \cdot \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \\
&= \left(a_1^{(3)} - y_1\right) \cdot \left[a_1^{(3)}\left(1 - a_1^{(3)}\right)\right] \cdot \Theta_{12}^{(2)} \cdot \left[a_2^{(2)}\left(1 - a_2^{(2)}\right)\right]
\end{aligned}$$

$$\begin{aligned}
\frac{\partial J_2(\Theta)}{\partial z_2^{(2)}} &= \frac{\partial J_2(\Theta)}{\partial a_2^{(3)}} \cdot \frac{\partial a_2^{(3)}}{\partial z_2^{(3)}} \cdot \frac{\partial z_2^{(3)}}{\partial a_2^{(2)}} \cdot \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \\
&= \left(a_2^{(3)} - y_2\right) \cdot \left[a_2^{(3)}\left(1 - a_2^{(3)}\right)\right] \cdot \Theta_{22}^{(2)} \cdot \left[a_2^{(2)}\left(1 - a_2^{(2)}\right)\right]
\end{aligned}$$

Example: Back Propagation

- Define

$$\delta^{(2)} := \begin{bmatrix} \dfrac{\partial J(\Theta)}{\partial z_1^{(2)}} \\ \\ \dfrac{\partial J(\Theta)}{\partial z_2^{(2)}} \end{bmatrix}$$

- Then

$$\begin{bmatrix} \dfrac{\partial J(\Theta)}{\partial \Theta_{10}^{(1)}}, \dfrac{\partial J(\Theta)}{\partial \Theta_{11}^{(1)}}, \dfrac{\partial J(\Theta)}{\partial \Theta_{12}^{(1)}} \\ \\ \dfrac{\partial J(\Theta)}{\partial \Theta_{20}^{(1)}}, \dfrac{\partial J(\Theta)}{\partial \Theta_{21}^{(1)}}, \dfrac{\partial J(\Theta)}{\partial \Theta_{22}^{(1)}} \end{bmatrix} = \delta^{(2)} \cdot \left[a_0^{(1)}, a_1^{(1)}, a_2^{(1)} \right]$$

- Recall that $a_0^{(1)} = a_0^{(2)} = 1$.
- This completes one iteration of the forward and back propagation.
- Continue this process until $\Theta_{ij}^{(l)}$ converge for $l = 1, 2$.

Example: Back Propagation

- How do we streamline the process of back propagation from $\delta^{(j+1)}$ to $\delta^{(j)}$?

Example: Back Propagation

- Note that

$$\left[\frac{\partial J(\Theta)}{\partial a_1^{(2)}}, \frac{\partial J(\Theta)}{\partial a_2^{(2)}}\right] = \left[\left(a_1^{(3)} - y_1\right) \cdot \left[a_1^{(3)}\left(1 - a_1^{(3)}\right)\right], \left(a_2^{(3)} - y_2\right) \cdot \left[a_2^{(3)}\left(1 - a_2^{(3)}\right)\right]\right] \begin{bmatrix} \Theta_{11}^{(2)} & \Theta_{12}^{(2)} \\ \Theta_{21}^{(2)} & \Theta_{22}^{(2)} \end{bmatrix}$$

$$= \left[\delta_1^{(3)}, \delta_2^{(3)}\right] \begin{bmatrix} \Theta_{11}^{(2)} & \Theta_{12}^{(2)} \\ \Theta_{21}^{(2)} & \Theta_{22}^{(2)} \end{bmatrix} = \left(\delta^{(3)}\right)^T \begin{bmatrix} \Theta_{11}^{(2)} & \Theta_{12}^{(2)} \\ \Theta_{21}^{(2)} & \Theta_{22}^{(2)} \end{bmatrix}$$

and

$$\left(\delta^{(2)}\right)^T = \left[\frac{\partial J(\Theta)}{\partial z_1^{(2)}}, \frac{\partial J(\Theta)}{\partial z_2^{(2)}}\right] = \left[\frac{\partial J(\Theta)}{\partial a_1^{(2)}}, \frac{\partial J(\Theta)}{\partial a_2^{(2)}}\right] \begin{bmatrix} \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} & 0 \\ 0 & \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \end{bmatrix}$$

$$= \left(\delta^{(3)}\right)^T \begin{bmatrix} \Theta_{11}^{(2)} & \Theta_{12}^{(2)} \\ \Theta_{21}^{(2)} & \Theta_{22}^{(2)} \end{bmatrix} \begin{bmatrix} \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} & 0 \\ 0 & \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \end{bmatrix}$$

Example: Back Propagation

- In general,

$$\left(\delta^{(j)}\right)^T = \left(\delta^{(j+1)}\right)^T \Theta^{(j)} \begin{bmatrix} \frac{\partial a_1^{(j)}}{\partial z_1^{(j)}} & 0 & \cdots & 0 \\ 0 & \frac{\partial a_2^{(j)}}{\partial z_2^{(j)}} & \cdots & 0 \\ \vdots & \vdots & \ldots & \vdots \\ 0 & 0 & \cdots & \frac{\partial a_{s_j}^{(j)}}{\partial z_{s_j}^{(j)}} \end{bmatrix}$$

where $\left(\delta^{(j)}\right)^T$ is of size $1 \times s_j$, $\left(\delta^{(j+1)}\right)^T$ is of size $1 \times s_{(j+1)}$, $\Theta^{(j)}$ is of size $s_{(j+1)} \times s_j$, and the diagonal matrix is of size $s_j \times s_j$.

Example: Back Propagation

- Back propagation starts with computing $\delta^{(L)}$.
- If neural network is used for regression, the total cost is

$$J(\Theta) = \frac{1}{2m} \sum_{i=1}^{m} \sum_{k=1}^{K} \left[\left(h_\theta(x^{(i)}) \right)_k - y_k^{(i)} \right]^2$$

  where m is the total number of samples and K is the total number of output units.

- For a specific sample, the associated cost is

$$J(\Theta) = \frac{1}{2} \sum_{k=1}^{K} \left[(h_\theta(x))_k - y_k \right]^2 = \frac{1}{2} \sum_{k=1}^{K} \left[a_k^{(L)} - y_k \right]^2$$

Example: Back Propagation

- $\delta^{(L)}$ can be computed as

$$\delta^{(L)} = \begin{bmatrix} \frac{\partial J(\Theta)}{\partial z_1^{(L)}} \\ \frac{\partial J(\Theta)}{\partial z_2^{(L)}} \\ \vdots \\ \frac{\partial J(\Theta)}{\partial z_{s_L}^{(L)}} \end{bmatrix} = \begin{bmatrix} \frac{\partial J(\Theta)}{\partial a_1^{(L)}} \cdot \frac{\partial a_1^{(L)})}{\partial z_1^{(L)}} \\ \frac{\partial J(\Theta)}{\partial a_2^{(L)}} \cdot \frac{\partial a_2^{(L)})}{\partial z_2^{(L)}} \\ \vdots \\ \frac{\partial J(\Theta)}{\partial a_{s_L}^{(L)}} \cdot \frac{\partial a_{s_L}^{(L)})}{\partial z_{s_L}^{(L)}} \end{bmatrix} = \begin{bmatrix} \frac{\partial J(\Theta)}{\partial a_1^{(L)}} \\ \frac{\partial J(\Theta)}{\partial a_2^{(L)}} \\ \vdots \\ \frac{\partial J(\Theta)}{\partial a_{s_L}^{(L)}} \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial a_1^{(L)}}{\partial z_1^{(L)}} & 0 & \cdots & 0 \\ 0 & \frac{\partial a_2^{(L)}}{\partial z_2^{(L)}} & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \frac{\partial a_{s_L}^{(L)}}{\partial z_{s_L}^{(L)}} \end{bmatrix}$$

$$= \begin{bmatrix} \left(a_1^{(L)} - y_1\right) \\ \left(a_2^{(L)} - y_2\right) \\ \vdots \\ \left(a_{s_L}^{(L)} - y_{s_L}\right) \end{bmatrix} \cdot \begin{bmatrix} a_1^{(L)}\left(1 - a_1^{(L)}\right) & 0 & \cdots & 0 \\ 0 & a_2^{(L)}\left(1 - a_2^{(L)}\right) & \cdots & 0 \\ \vdots & & & \\ 0 & 0 & \cdots & a_{s_L}^{(L)}\left(1 - a_{s_L}^{(L)}\right) \end{bmatrix}$$

Example: Back Propagation

- If a neural network is used for classification, the total cost is

$$J(\Theta) = -\frac{1}{m}\left\{\sum_{I=1}^{m}\sum_{k=1}^{K}\left[y_k^{(i)}\log\left(h_\Theta(x^{(i)})\right)_k + \left(1-y_k^{(i)}\right)\log\left(1-\left(h_\Theta(x^{(i)})\right)_k\right)\right]\right\}$$

  where m is the total number of samples and K is the total number of output units.

- For a specific sample, the associated cost is

$$\begin{aligned} J(\Theta) &= -\sum_{k=1}^{K}\left[y_k\log\left(h_\Theta(x)\right)_k + (1-y_k)\log\left(1-\left(h_\Theta(x)\right)_k\right)\right] \\ &= -\sum_{k=1}^{K}\left[y_k\log a_k^{(L)} + (1-y_k)\log\left(1-a_k^{(L)}\right)\right] \end{aligned}$$

Example: Back Propagation

- $\delta^{(L)}$ can be computed as

$$\delta^{(L)} = \begin{bmatrix} \frac{\partial J(\Theta)}{\partial z_1^{(L)}} \\ \frac{\partial J(\Theta)}{\partial z_2^{(L)}} \\ \vdots \\ \frac{\partial J(\Theta)}{\partial z_{s_L}^{(L)}} \end{bmatrix} = \begin{bmatrix} \frac{\partial J(\Theta)}{\partial a_1^{(L)}} \cdot \frac{\partial a_1^{(L)})}{\partial z_1^{(L)}} \\ \frac{\partial J(\Theta)}{\partial a_2^{(L)}} \cdot \frac{\partial a_2^{(L)})}{\partial z_2^{(L)}} \\ \vdots \\ \frac{\partial J(\Theta)}{\partial a_{s_L}^{(L)}} \cdot \frac{\partial a_{s_L}^{(L)})}{\partial z_{s_L}^{(L)}} \end{bmatrix} = \begin{bmatrix} \frac{\partial J(\Theta)}{\partial a_1^{(L)}} \\ \frac{\partial J(\Theta)}{\partial a_2^{(L)}} \\ \vdots \\ \frac{\partial J(\Theta)}{\partial a_{s_L}^{(L)}} \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial a_1^{(L)}}{\partial z_1^{(L)}} & 0 & \cdots & 0 \\ 0 & \frac{\partial a_2^{(L)}}{\partial z_2^{(L)}} & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \frac{\partial a_{s_L}^{(L)}}{\partial z_{s_L}^{(L)}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\left(a_1^{(L)} - y_1\right)}{a_1^{(L)}\left(1 - a_1^{(L)}\right)} \\ \frac{\left(a_2^{(L)} - y_2\right)}{a_2^{(L)}\left(1 - a_2^{(L)}\right)} \\ \vdots \\ \frac{\left(a_{s_L}^{(L)} - y_{s_L}\right)}{a_{s_L}^{(L)}\left(1 - a_{s_L}^{(L)}\right)} \end{bmatrix} \cdot \begin{bmatrix} a_1^{(L)}\left(1 - a_1^{(L)}\right) & 0 & \cdots & 0 \\ 0 & a_2^{(L)}\left(1 - a_2^{(L)}\right) & \cdots & 0 \\ \vdots & & & \\ 0 & 0 & \cdots & a_{s_L}^{(L)}\left(1 - a_{s_L}^{(L)}\right) \end{bmatrix} = \begin{bmatrix} a_1^{(L)} - y_1 \\ a_2^{(L)} - y_2 \\ \vdots \\ a_{s_L}^{(L)} - y_{s_L} \end{bmatrix}$$

Gradient computation for one iteration

- Given training samples $\left\{\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \cdots, \left(x^{(m)}, y^{(m)}\right)\right\}$
- Denote $D_{i,j}^{(l)} := \dfrac{\partial J(\Theta)}{\partial \Theta_{i,j}^{(l)}}$. Gradient for one iteration is computed as follows:

Set $\Delta_{i,j}^{(l)} = 0$ for all l, i, j.

for $i = 1$ to m {

Set $a^{(1)} = x^{(i)}$ and $a^{(L)} = y^{(i)}$.

Perform forward propagation to compute $a^{(l)}$ for $l = 2, 3, \cdots, L$.

Compute $\delta^{(L)}$ according to the total cost function.

Compute $\delta^{(L-1)}, \delta^{(L-2)}, \cdots, \delta^{(2)}$ using back propagation.

Compute $\Delta_{i,j}^{(l)} := \Delta_{i,j}^{(l)} + a_j^{(l)} \delta_i^{(l+1)}$ OR $\Delta^{(l)} := \Delta^{(l)} + \delta^{(l+1)} \left(a^{(l)}\right)^T$

}

$D^{(l)} := \dfrac{1}{m} \Delta^{(l)}$ and $\dfrac{\partial J(\Theta)}{\partial \Theta_{ij}^{(l)}} = D_{ij}^{(l)}$

Overall gradient descent algorithm

- Given learning rate α and training samples $\left\{\left(x^{(1)}, y^{(1)}\right), \left(x^{(2)}, y^{(2)}\right), \cdots, \left(x^{(m)}, y^{(m)}\right)\right\}$
- Denote $D_{i,j}^{(l)} := \dfrac{\partial J(\Theta)}{\partial \Theta_{i,j}^{(l)}}$. Gradient descent proceeds as follows:

for index_iter = 1 to total_number_of_iterations {

Set $\Delta_{i,j}^{(l)} = 0$ for all l, i, j.

for $i = 1$ to m {

Set $a^{(1)} = x^{(i)}$ and $a^{(L)} = y^{(i)}$.

Perform forward propagation to compute $a^{(l)}$ for $l = 2, 3, \cdots, L$.

Compute $\delta^{(L)}$ according to the total cost function.

Compute $\delta^{(L-1)}, \delta^{(L-2)}, \cdots, \delta^{(2)}$ using back propagation.

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}

$D^{(l)} := \dfrac{1}{m} \Delta^{(l)}$ and $\dfrac{\partial J(\Theta)}{\partial \Theta_{ij}^{(l)}} = D_{ij}^{(l)}$

$\Theta^{(l)} := \Theta^{(l)} - \alpha D^{(l)}$

}

Parameter Initialization

- Before training begins, parameters that will be learned must be assigned starting values.
- Training then adjusts them using gradient descent.
- We cannot initialize all parameters to zero. Otherwise
  - All neurons compute identical outputs.
  - all gradients remain identical.
  - neurons never learn.
  - This is called the **symmetry problem**.
- Thus parameters must be initialized randomly.
- Simple random initialization using a normal distribution:

$$\theta \sim \mathcal{N}(0, \sigma^2)$$

  or uniform distribution:

$$\theta \sim \mathcal{U}(-a, a)$$

Strategies to Update Parameters

Batch gradient descent (Full GD)

- Use all data to compute gradient
- Intuition: Look at the entire dataset before making a decision.
- Characteristics
  - Very accurate gradient
  - Smooth and stable updates
  - Deterministic
- Limitations:
  - Slow for large datasets
  - High memory usage
  - Cannot scale well

Strategies to Update Parameters

Stochastic gradient descent (SGD)

- Use one sample at a time
- Intuition: Learn from one sample at a time.
- Characteristics
  - Very fast update
  - Low memory requirement
  - Introduce randomness
- Limitations:
  - Very noisy update
  - Cost curve fluctuates
  - May not converge smoothly

Strategies to Update Parameters

Mini-Batch gradient descent (Most important)

- Use a small batch of data
- Intuition: Learn from a small subset of the samples at a time.
- Characteristics
  - balance between speed and stability
  - efficient on modern hardware (GPU)
  - moderately noisy updates
- Limitations:
  - still requires tuning batch size
  - some randomness remains

Strategies to Update Parameters

Effect on optimization path

- Batch GD: smooth path and straight descent
- SGD: zig-zag path and highly irregular
- Mini-Batch: slightly noisy but stable

Effect of batch size in Mini-Batch

- smaller batch: $\rightarrow$ more noise $\rightarrow$ better exploration
- larger batch: $\rightarrow$ smoother $\rightarrow$ faster convergence
- Rule of thumb:
  - small datasets $\rightarrow$ small batch
  - large datasets $\rightarrow$ larger batch

Epoch

- An **epoch** is defined as one complete pass of the neural network through the entire training dataset.
- Neural networks usually do not learn enough from one pass and thus model needs many repeated exposures to the data.
- Example:
  - Epoch 1: model very inaccurate
  - Epoch 20: model starts capturing patterns
  - Epoch 100: model converges

Analogy: like studying a textbook - reading once is rarely enough.

Epoch and Parameter Update

- Batch gradient descent: one epoch = one parameter update because gradient uses entire dataset.
- Stochastic gradient descent: one epoch = m parameter update because model updates after every sample.
- Mini-batch gradient descent: For a dataset size = 200 and batch size = 20, then one epoch = 200/20=10 parameter updates.

Thank you!